



TM980

OPC UA basics and use

Requirements

Basic knowledge	Basic technical understanding
Training module	TM210 - Automation Studio basics
Software	UaExpert 1.6.3 Automation Studio 4.12 OPC UA FX package 1.2.0
Hardware	X20CP1586 (ETAL210 - ETA light system control technology) X20BC008U (ETAL980 - ETA Light OPC UA)

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1 Introduction

OPC Unified Architecture (OPC UA) is a manufacturer-independent communication protocol for automation applications in industry. It enables end-to-end communication from individual sensors and actuators to the ERP system or the cloud.

B&R has been relying on OPC UA for years and is an active member of various OPC Foundation working groups. Find out more about the solutions and products B&R offers with OPC UA.

Configuring OPC UA devices and assigning access rights in B&R's Automation Studio engineering environment is completed with just a few clicks. PLCopen function blocks are used to implement OPC UA functions in an automation project.



Figure 1: OPC UA logo

1.1 Learning objectives

With selected application examples and exercises, participants will learn the basics of OPC UA communication technology.

- Participants will learn the basics of OPC UA technology.
- Participants will become familiar with the possibilities, advantages and typical use cases of OPC UA.
- Participants will be able to connect to an OPC UA server via an OPC UA client.
- Participants will be able to configure an OPC UA client in AS.
- Participants will be able to commission the Automation Runtime OPC UA server in Automation Studio.
- Participants will be able to configure the user role system in AS.
- Participants are familiar with the advantages of publish/subscribe architecture and can configure it in Automation Studio.
- Participants will learn how to use the OPC UA library in Automation Studio.
- Participants will be able to implement and call OPC UA methods.
- Participants will learn how to handle certificates.

2 OPC UA basics

2.1 OPC Unified Architecture

Open Platform Communications Unified Architecture (OPC UA) and is an international standard for secure, reliable, manufacturer- and platform-independent information exchange in industrial communication.

OPC UA enables integrated communication ranging from ERP systems to individual sensors and actuators and makes semantic interoperability available in automation industry. OPC UA is flexible and fully independent and contributes substantially to the success of the 4th industrial (r)evolution.

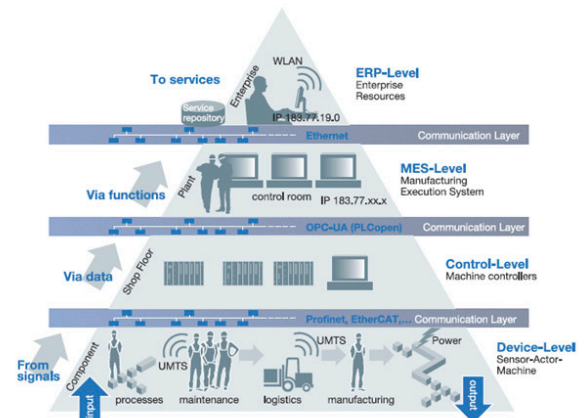


Figure 2: OPC UA pyramid

Advantages

- International standard (IEC 62541)
- Safe data exchange
- Reliable
- Vendor-independent
- Platform-independent
- Vertical and horizontal communication

2.1.1 International standard (IEC 62541)

OPC UA has had specifications defined by the OPC Foundation and has been standardized in IEC 62541.

The OPC UA specification is composed of 14 parts and is continuously supplemented by specifications from other organizations such as PLCopen and other consortia.

Standard	OPC UA specifications
IEC/TR 62541-1	Part 1: Overview and concepts
IEC/TR 62541-2	Part 2: Security model
IEC 62541-3	Part 3: Address space model
IEC 62541-4	Part 4: Services
IEC 62541-5	Part 5: Information model
IEC 62541-6	Part 6: Mapping
IEC 62541-7	Part 7: Profiles
IEC 62541-8	Part 8: Data access
IEC 62541-9	Part 9: Alarms and conditions
IEC 62541-10	Part 10: Programs
IEC 62541-11	Part 11: Historical access
	Part 12: Discovery
IEC 62541-13	Part 13: Aggregates
	Part 14: PubSub
IEC 62541-100	PLCopen IEC 61131-3

Table 1: OPC UA specifications

OPC Foundation

The OPC Foundation is an independent committee that specifies and further develops the OPC UA standard.

B&R has been relying on the OPC UA standard for years and is an active member of various OPC Foundation working groups.

See <https://opcfoundation.org/>



Figure 3: OPC Foundation

2.2 Use cases

OPC UA is used in various constellations on B&R systems:

- For connecting HMI applications with controllers, e.g. mapp View
- For data exchange between B&R controllers
- For data exchange with other controllers
- For connecting to fieldbus devices
(B&R bus controller, OPC UA capable sensors or actuators)

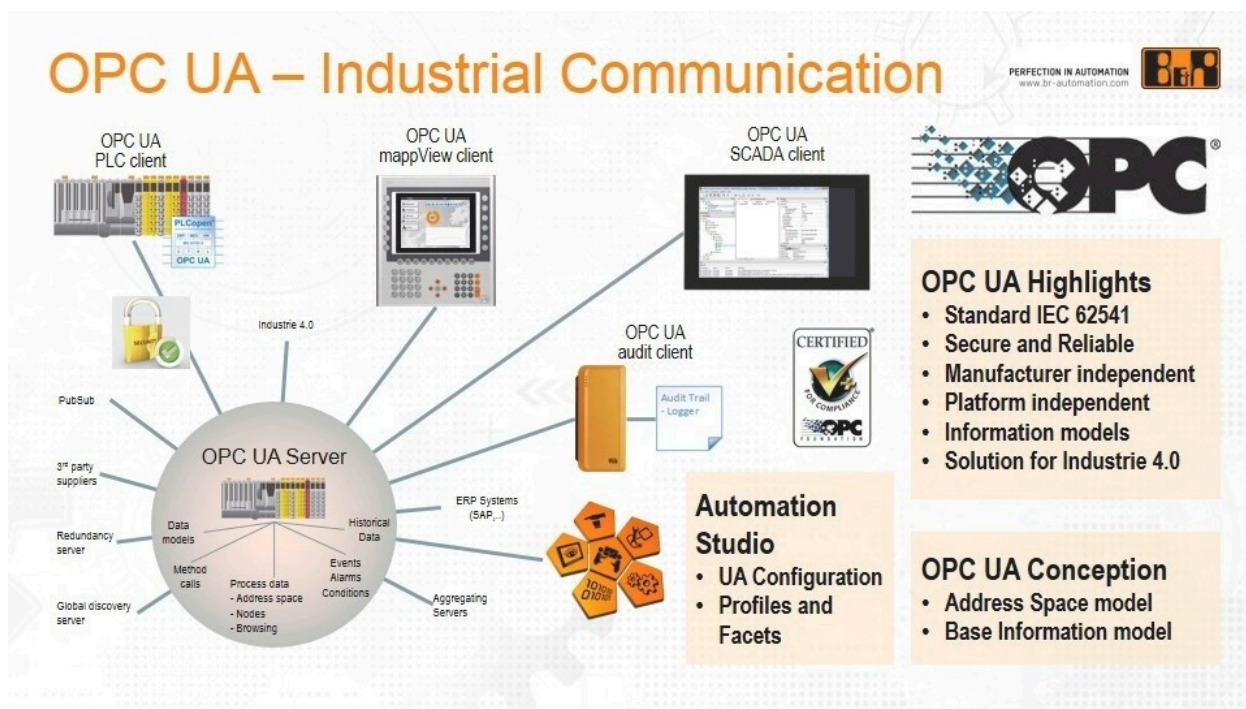


Figure 4: OPC UA - Industrial communication



2.2.1 HMI applications

A more common and typical application of OPC UA is exchanging process variables between HMI applications and PLC systems.

Explained in simple terms, HMI applications as OPC UA clients connect with PLC systems as OPC UA servers and read or write OPC UA node attributes in the OPC UA address space. OPC UA nodes receive process variable data to/from the PLC via the OPC UA server.

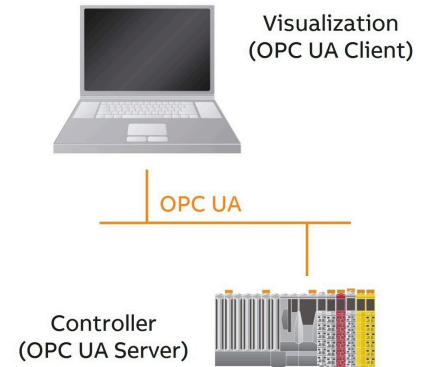


Figure 5: Data exchange between HMI application and control system

Monitored items vs. continuous reading

In this case, HMI applications should prevent unnecessary network traffic caused by continuous reading of the node. Any nodes needed by the HMI application are subscribed to the server and are tasked with monitoring for value modifications (MonitoredItems). If node values change, the clients are informed (Publish) and only the changed values are transferred.

mapp View

The mapp View HMI system has been developed by B&R and uses OPC UA for communication to PLC systems from B&R or other manufacturers.

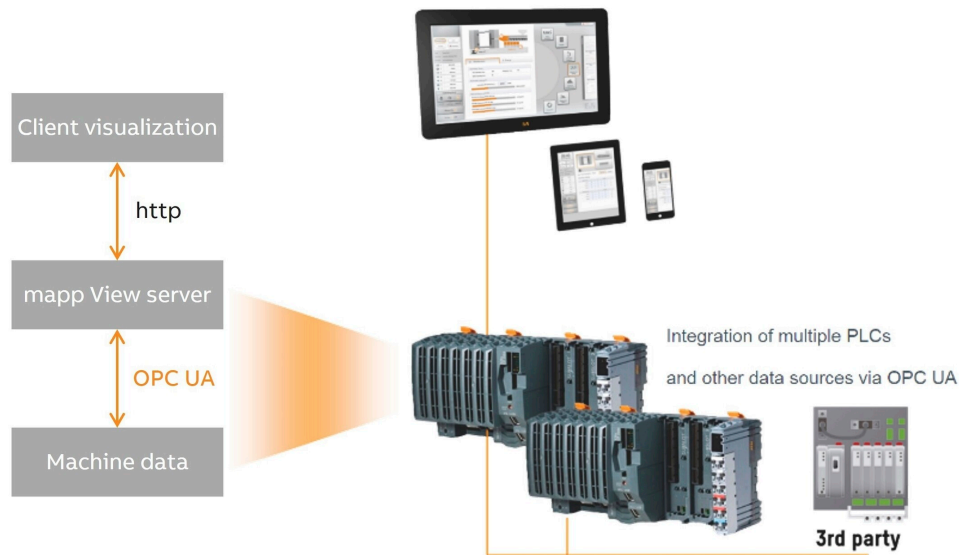


Figure 6: How mapp View works

2.2.2 OPC UA bus controller

This bus controller provides OPC UA server functions. This allows any OPC UA clients access to read or write data from I/O modules connected to the bus controller.

For additional information about using a bus controller, see chapter 3 "[OPC UA bus controller integration](#)" on page 12.

2.2.3 Controller to controller

With OPC UA, communication between PLC systems can also be facilitated and harmonized. For this, the function blocks specified by PLCopen are used.

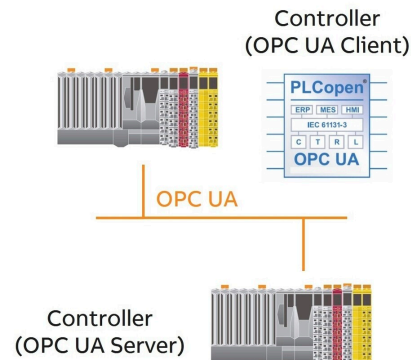


Figure 7: Connection between OPC UA client and OPC UA server

There are more than 20 function blocks available in Automation Studio library AsOpcUac that can be used to handle establishing connections, transferring data or calling methods.

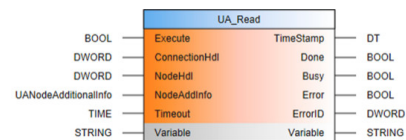


Figure 8: Function block "UA_Read"

2.2.4 Audit clients

Audit mechanisms are included on the B&R OPC UA server. These inform an audit client which user changed which OPC UA node as well as what they used to make the change and when it occurred.

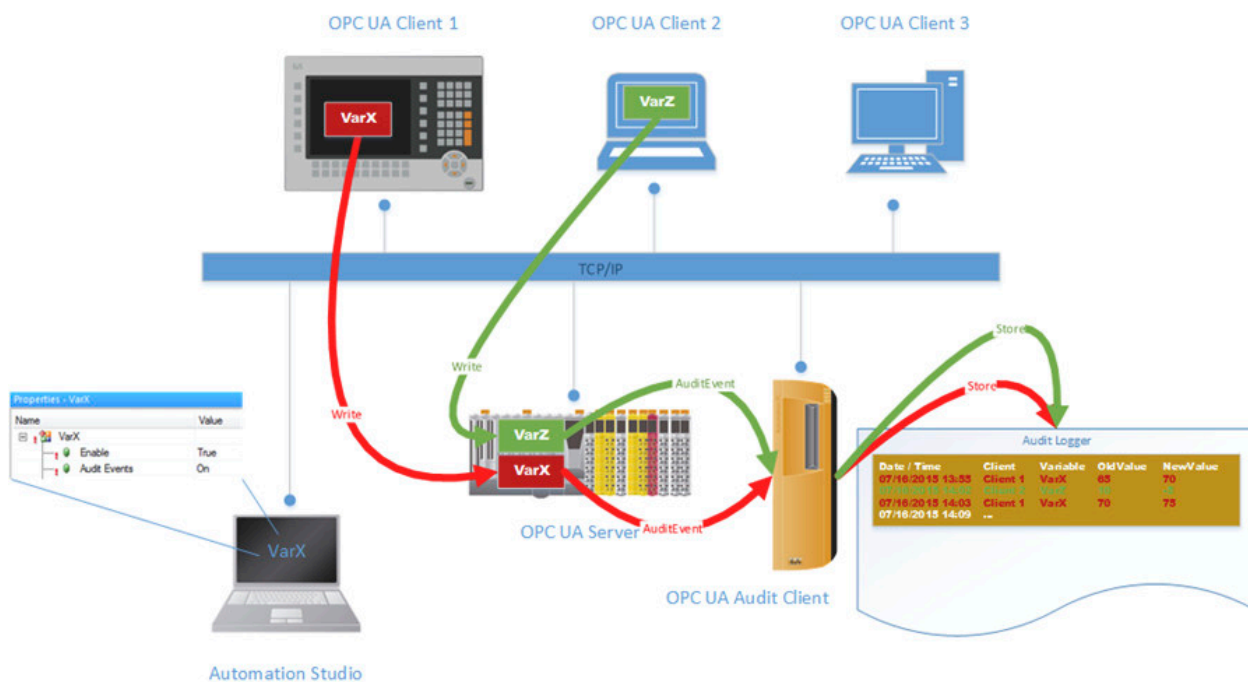


Figure 9: Audit clients operating principle

For example, the OPC UA audit client can be informed by the OPC UA server via audit event that OPC UA client 1 changed variable VarX from 65 to 70 degrees at 1:55 PM. The nodes that should trigger audit events are set in the Automation Studio configuration.

2.2.5 Third-party systems

OPC UA third-party systems can also be connected to B&R systems without any programming effort. Just like for the OPC UA bus controller, the third-party device is configured in Automation Studio. The required nodes can be declared as channels.

With this configuration, the OPC UA can set up drivers in Automation Runtime on the third-party device as OPC UA client appropriate monitored items and transfer them as inputs to the PLC process variables or even write the process variables of the PLC as outputs onto the third-party device using the OPC UA Write service.

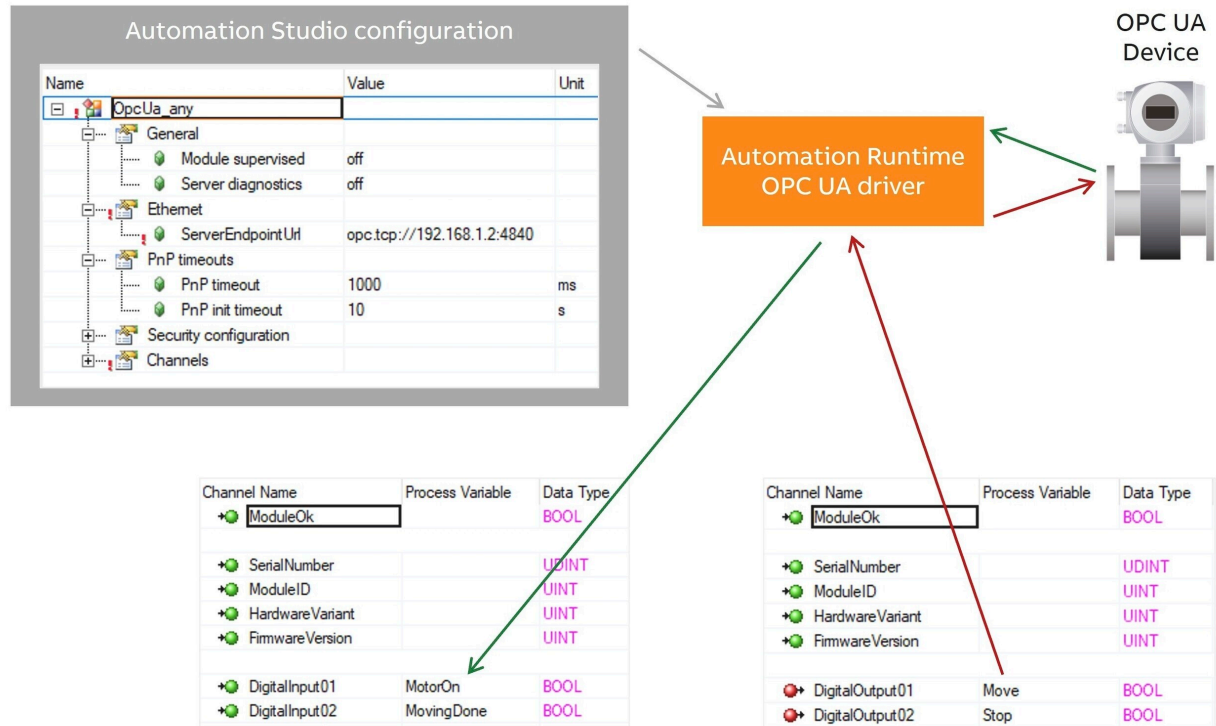


Figure 10: Integration of 3rd-party systems

2.3 Information model

On the B&R OPC UA server, the PLC system is mapped in the OPC UA address space based on the OPC UA specifications.

- Part 3 (Address Space Model)
- Part 5 (Information model)
- Companion Specification PLCopen OPC UA Information Model 1.00 Specification

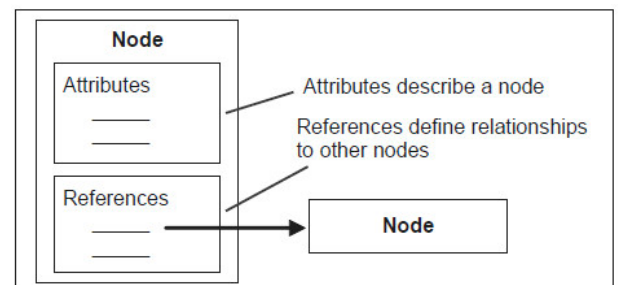


Figure 11: Information model

The address space is defined using nodes and their attributes, which are connected using references.

2.3.1 OPC UA nodes

The B&R OPC UA server contains OPC UA nodes that are required according to the specifications (e.g. the server object). In addition, it contains all nodes that are specifically created on the B&R system during startup or during operation of the B&R OPC UA server (e.g. tasks and variables of the tasks), including their type descriptions, which are mapped as nodes.

NodeID

A node is uniquely identified in the address space via its NodeID. The NodeID consists of the NamespaceIndex, Identifier and IdentifierType. There is also the IdentifierType GUID, which is not used by B&R.

- **NamespaceIndex**
This regulates the responsibility for identifying a node, see "[Namespaces](#)" on page 10.
- **Identifier**
This consists either of a numeric value (resource-saving in small systems) [e.g. 2253 (server object)] or a string [e.g. "Submarine::ballast"].
- **IdentifierType** (numeric or string)
Specifies whether the identifier is defined by a numerical value or by a character string.

Browse name

- If it is not known which nodes a server contains, then the address space can be searched with browser services (browse) starting from the root node or other known nodes. When browsing, a node is identified using the BrowseName.

2.3.2 Namespaces

Since the nodes are defined by the OPC Foundation, the PLCopen organization or B&R itself, two identical identifiers could be assigned at random, for example.

In order to prevent this, the responsibility of the nodes in the OPC UA address space is regulated using the namespace. The namespace therefore specifies which institution has defined the node (naming authority) and is listed in the form of a namespace URI.

Automation Runtime:

Namespace URI	Description
http://opcfoundation.org/UA/	Types and objects specified by the OPC Foundation
<a href="urn:<hostname>/BR/UA/EmbeddedServer">urn:<hostname>/BR/UA/EmbeddedServer	Types and objects for server provider
http://opcfoundation.org/UA/DI/	Types and objects for device integration
http://PLCopen.org/OpcUa/IEC61131-3/	Types and objects for PLCopen
urn:B&R/plc/	Static B&R types and objects
urn:B&R/pv/	B&R information model for PLC process variables
urn:PLCopen/pv/	PLCopen information model for PLC process variables
http://br-automation.com/Diagnostics	Types and objects for Commissioning Cockpit diagnostics

Table 2: Automation Runtime namespaces



[Communication \ OPC UA \ Information model \ OPC UA nodes](#)

OPC UA bus controller

Namespace URI	Description
http://opcfoundation.org/UA/	Types and objects specified by the OPC Foundation
http://www.br-automation.com/io-system/	B&R Information model for configuration and process variables

Table 3: OPC UA bus controller namespaces



[Hardware \ X20 System \ X20 module \ Bus controller \ X20BC008U \ Bus controller information model](#)

PowerPanel T-Series

Namespace URI	Description
http://opcfoundation.org/UA/	Types and objects specified by the OPC Foundation
urn:<hostname>/BR/UA/EmbeddedServer	Types and objects for server provider
http://opcfoundation.org/UA/DI/	Types and objects for device integration
http://br-automation.com/OpcUa/BrTypes/	General types and objects defined by B&R
http://br-automation.com/OpcUa/HMI/Terminal/	Device types and objects defined by B&R

Table 4: PowerPanel T-Series namespaces



[Hardware \ Power Panel \ Power Panel T50 \ Software \ OPC UA server \ Information model](#)

Since the URI tends to be quite long and specifying it in the NodeID is not desired, these URIs are managed in a "NamespaceArray" in the server object. Only the index in this table is used in the NodeID.

- NamespaceIndex 0: http://opcfoundation.org/UA/
- NamespaceIndex 3: http://opcfoundation.org/UA/DI/
- NamespaceIndex 6: urn:B&R/pv/

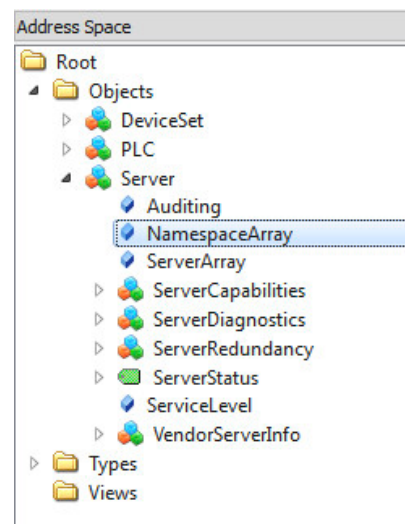


Figure 12: Address space - NamespaceArray

3 OPC UA bus controller integration

The bus controller X20BC008U connects OPC UA systems with the X20 I/O system. In a corresponding information model, which is derived from the OPC UA DI model, data from both the bus controller and the X20 I/O modules can be requested or transmitted using OPC UA services. The OPC UA standard means that the bus controller can be used not only by B&R, but also by any OPC UA client.

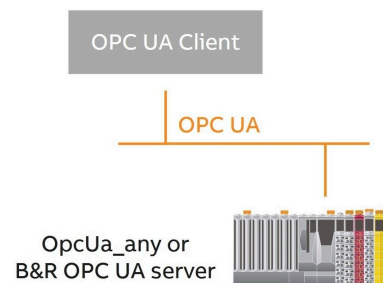


Figure 13: Bus controller communication

Once the hardware configuration has been transferred to the B&R controller, the corresponding OPC UA driver can communicate with the X20BC008U bus controller in Automation Runtime via the standard Ethernet interface using OPC UA.

3.1 Preparation and configuration of the bus controller

In order to be able to use the bus controller, the first step is to prepare the network in which it is to be used. It is possible to either use a local network with fixed PC and bus controller IP addresses or assign the IP addresses within a DHCP network and use the hostnames.

If the topology and network have been set up, the bus controller can now be configured. Network settings can be made using the address switches on the front of the bus controller. It is possible to set a fixed IP address or to get the IP address from a DHCP server.



[Hardware \ X20 system \ X20 modules \ Bus controller \ X20BC008U \ Address switch](#)

If the bus controller is addressable in the network, it is possible to update the firmware. The latest firmware is available for download on the B&R website. This can then be loaded onto the bus controller by accessing the IP address via a web browser. For more information, see ["Optional exercise: Perform a firmware update for X20BC008U" on page 50.](#)

Exercise: Build the topology and adjust the network settings

In this exercise, the network topology is built and the PC's network settings are adjusted. It is possible to set up a local network either using the integrated hub of the bus controller or to set up a network via a DHCP server.

Local network:

- 1) Build the network topology according to the sketch.
- 2) PC network settings:

IP address: 192.168.1.128
Subnet mask: 255.255.255.0

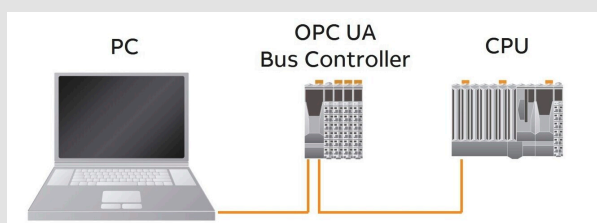


Figure 14: Structure of local network

Network via DHCP server:

- 1) Build the network topology according to the sketch.
- 2) Make sure that the IP address of the PC is assigned automatically.

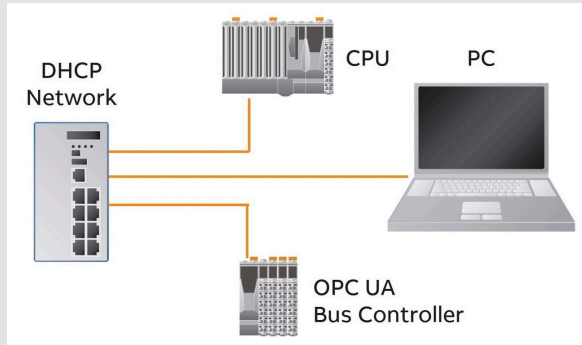


Figure 15: Network setup via DHCP server

Exercise: Configure the OPC UA bus controller

The objective of this exercise is to configure the OPC UA bus controller X20BC008U according to the network topology. To do this, turn the address switches to a defined position. The positions for the different modes are described in Automation Help.

- 1) Static IP address for the bus controller:
Set the address switch to a value between "0x01 - 0x7F".
- 2) DHCP mode for the bus controller:
Set the address switch to a value between "0x80 - 0xEF".

3.2 Establishing a connection with Unified Automation UaExpert client

The free program "UaExpert", which was developed by Unified Automation, is used to establish the connection to the bus controller. It is a simple and versatile OPC UA client.

In order to use this program, you must register on the website.



<https://www.unified-automation.com/downloads/opc-ua-clients.html>

For detailed instructions about how to use the UaExpert client, see the UaExpert help documentation in chapter "First Steps".

1) Add a server.

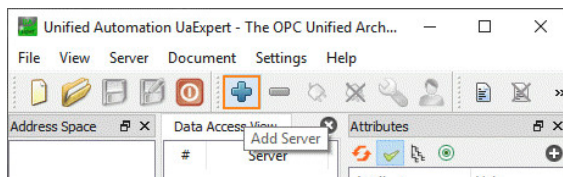


Figure 16: UaExpert: Add a new server via the toolbar

2) Enter the server's URL.

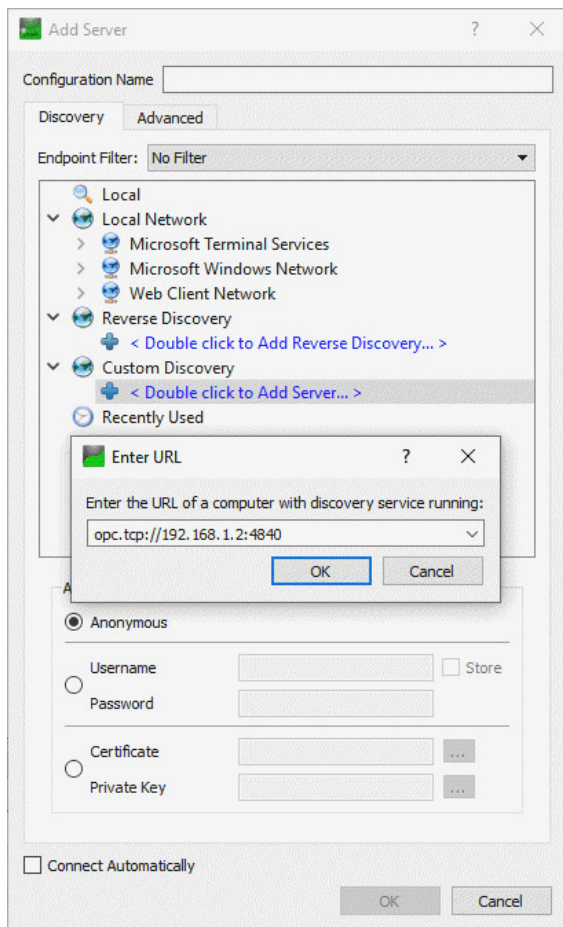


Figure 17: UaExpert: Enter server URL

3) Select signature and encryption.

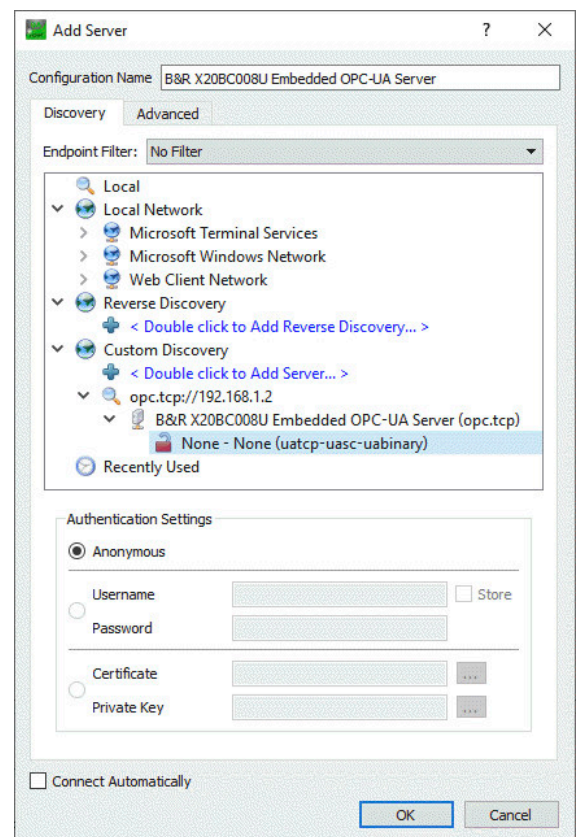


Figure 18: UaExpert: Select signature and encryption

4) Connect to the server

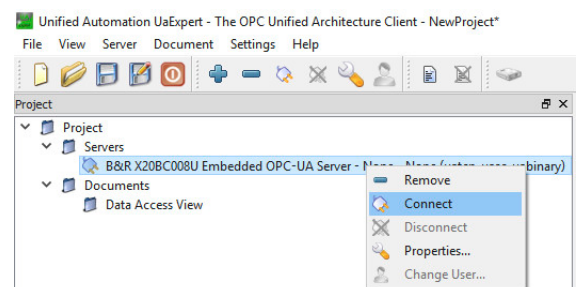


Figure 19: UaExpert: Connect the server



If the UaExpert certificate is not in the server's trusted list (6.1 "Trusted list" on page 37), a connection cannot be established between the UaExpert client and the OPC UA server.

Log			
Timestamp	Source	Server	Message
04.06.2019 09:33:11.419	Server Node	B&R Embedded OPC-UA Server and Client	Endpoint: 'opc.tcp://br-automation:4840'
04.06.2019 09:33:11.419	Server Node	B&R Embedded OPC-UA Server and Client	Security policy: 'http://opcfoundation.org/UA/SecurityPolicy#Basic256'
04.06.2019 09:33:11.419	Server Node	B&R Embedded OPC-UA Server and Client	ApplicationUri: 'urn:br-automation/BR/UA/EmbeddedServer'
04.06.2019 09:33:11.419	Server Node	B&R Embedded OPC-UA Server and Client	Used UserTokenType: Anonymous
04.06.2019 09:33:11.465	Server Node	B&R Embedded OPC-UA Server and Client	Error 'BadSecurityChecksFailed' was returned during OpenSecureChannel
04.06.2019 09:33:11.465	Server Node	B&R Embedded OPC-UA Server and Client	Connection status of server 'B&R Embedded OPC-UA Server and Client' changed to 'Disconnected'.

Figure 20: Result: Trusted list



[Hardware \ X20 system \ X20 modules \ Bus controller \ X20BC008U \ Connect to UaExpert](#)

Exercise: Establish a connection with the UaExpert bus controller

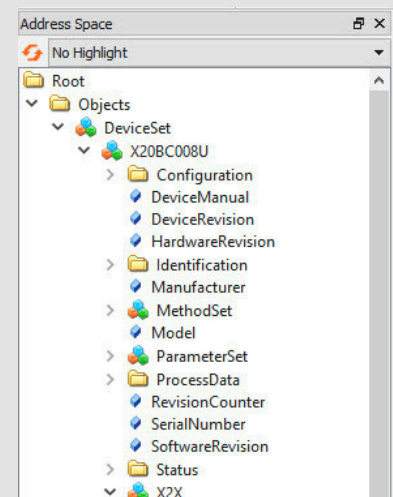
Program "UaExpert" is used to establish a connection to bus controller X20BC008U. Automation Help provides information about how to establish a connection.

- 1) Install UaExpert.
- 2) Open UaExpert.
- 3) Add a new server.
- 4) Connect to the server.



As soon as the connection has been established, the structure of the address space provided by the bus controller will be displayed in its own window.

The bus controller is shown in subsection "DeviceSet" and contains all objects provided by the bus controller.



3.3 Reading data

When the connection is established, it is possible to access the desired data in the "Address space" window. The following path can be used, for example, to navigate to the inputs of the DI module:

Root / Objects / DeviceSet / X20BC008U / X2X / SubDevices / ST002| X20DI6371 / ProcessData

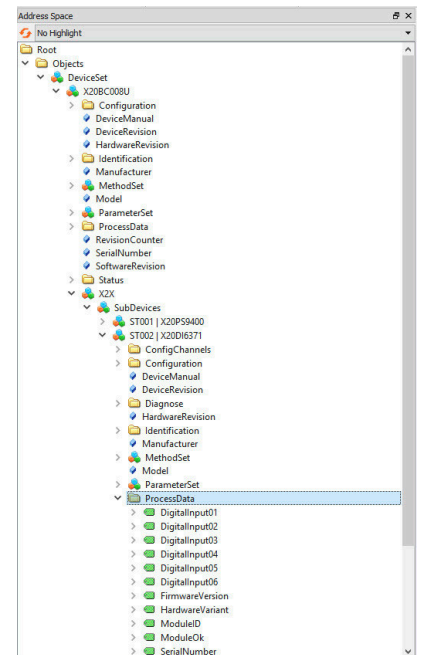


Figure 21: UaExpert: Address space

The inputs or nodes can then be dragged and dropped into the "Data access view".

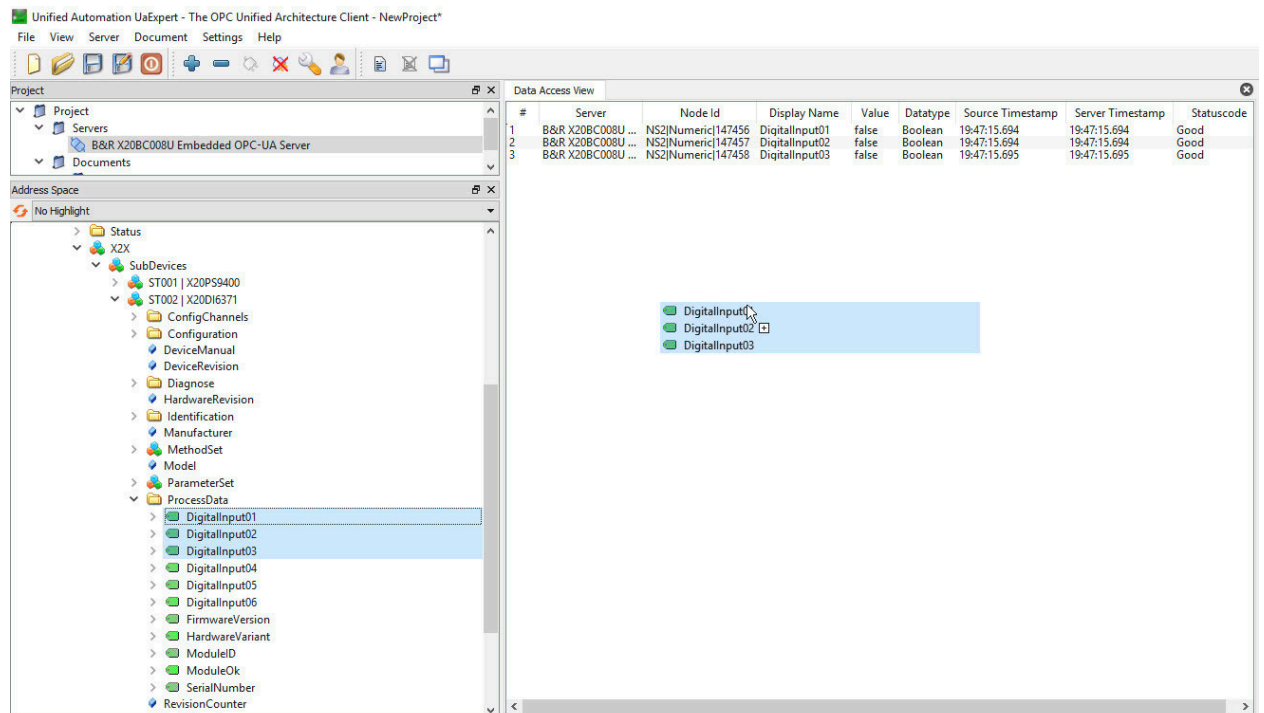


Figure 22: UaExpert: Adding nodes via drag-and-drop

The inputs and outputs or nodes can now be read out or set. For the latter, double-click on the corresponding output and select the check box.

Data Access View								
#	Server	Node Id	Display Name	Value	Datatype	Source Timestamp	Server Timestamp	Statuscode
1	B&R X20BC008U ...	NS2 Numeric 147456	DigitalInput01	☒	Boolean	20:05:44.413	20:05:44.413	Good
2	B&R X20BC008U ...	NS2 Numeric 147457	DigitalInput02	false	Boolean	19:47:15.694	19:47:15.694	Good
3	B&R X20BC008U ...	NS2 Numeric 147458	DigitalInput03	false	Boolean	19:47:15.695	19:47:15.695	Good
4	B&R X20BC008U ...	NS2 Numeric 212992	DigitalOutput01	true	Boolean	20:05:42.163	20:05:42.163	Good
5	B&R X20BC008U ...	NS2 Numeric 212993	DigitalOutput02	false	Boolean	20:05:40.913	20:05:40.913	Good
6	B&R X20BC008U ...	NS2 Numeric 212994	DigitalOutput03	true	Boolean	20:05:42.913	20:05:42.913	Good
7	B&R X20BC008U ...	NS2 Numeric 278528	AnalogInput01	859	Int16	20:06:07.663	20:06:07.663	Good

Figure 23: UaExpert: Reading and setting nodes

Exercise: Read inputs / Set outputs

The objective of this exercise is to read and set variables from the bus controller using the UaExpert program.

- 1) Navigate to the following path in window "Address space":

Root / Objects / DeviceSet / X20BC008U / X2X / SubDevices / ST002|X20DI6371 / ProcessData

- 2) Drag nodes "DigitalInput1" and "DigitalInput2" into "Data access view".

- 3) Navigate to the following path in window "Address space":

Root / Objects / DeviceSet / X20BC008U / X2X / SubDevices / ST003|X20DO6322 / ProcessData

- 4) Drag nodes "DigitalOutput1" and "DigitalOutput2" into "Data access view".

- 5) Navigate to the following path in window "Address space":

Root / Objects / DeviceSet / X20BC008U / X2X / SubDevices / ST004|X20AI2222 / ProcessData

- 6) Drag node "AnalogInput1" into the "Data Access View".

- 7) Observe the value change in "Data Access View" if the switches or buttons of the [OPC UA ETA light](#) are enabled or the potentiometer is changed.

- 8) Observe the status change of the green LED on the [OPC UA ETA light](#) if the digital outputs are set via "Data Access View".



Value changes can be performed and observed in "Data Access View".

Data Access View								
#	Server	Node Id	Display Name	Value	Datatype	Source Timestamp	Server Timestamp	Statuscode
1	B&R X20BC008U ...	NS2 Numeric 1...	DigitalInput01	true	Boolean	04:11:53.710	04:11:53.710	Good
2	B&R X20BC008U ...	NS2 Numeric 1...	DigitalInput02	false	Boolean	04:04:57.210	04:04:57.210	Good
3	B&R X20BC008U ...	NS2 Numeric 1...	DigitalInput03	false	Boolean	04:05:01.210	04:05:01.210	Good
4	B&R X20BC008U ...	NS2 Numeric 2...	DigitalOutput01	true	Boolean	04:11:57.210	04:11:57.210	Good
5	B&R X20BC008U ...	NS2 Numeric 2...	DigitalOutput02	false	Boolean	04:11:46.960	04:11:46.960	Good
6	B&R X20BC008U ...	NS2 Numeric 2...	DigitalOutput03	false	Boolean	04:11:49.460	04:11:49.460	Good
7	B&R X20BC008U ...	NS2 Numeric 2...	InputRegister0	10652	Int16	04:13:04.460	04:13:04.460	Good

Figure 24: UaExpert: Result: Reading inputs and outputs

Optional exercise: Change the "packed" format using a method call

The objective of this exercise is to configure the "packed" format on the I/O module (X20DI6371) via the "Packed inputs" node. The default setting for the format of the digital I/Os is "unpacked" (0 for off).

- 1) In the "Address space" window, navigate to the following path and check whether DigitalInput01 to DigitalInput06 are listed:

Root / Objects / DeviceSet / X20BC008U / X2X / SubDevices / ST002 | X20DI6371 / ProcessData

- 2) Navigate to the following path in window "Address space":

Root / Objects / DeviceSet / X20BC008U / X2X / SubDevices / ST002 | X20DI6371 / Configuration / General / Packed inputs

- 3) Use drag & drop to place the "Packed inputs" note in the "Data Access View".

- 4) Set the value of "Packed inputs" to 1 to allow the configuration.

- 5) Navigate to the following path in window "Address space":

Root / Objects / DeviceSet / X20BC008U / X2X / SubDevices / ST002 | X20DI6371 / Configuration / Control / Apply changes

- 6) Right-click on the "ApplyChanges" method and click "Call" to apply the settings.

- 7) Click on "Rebrowse" in the "Address space" window to update the address space.



In the "Data Access View", all 6 digital inputs are displayed bit-coded under the DigitalInput (byte) node.

#	Server	Node Id	Display Name	Value	Datatype	Source Timestamp	Server Timestamp	Statuscode
1	B&R X20BC008U	NS2/Numeric1...	Packed inputs	1 (on)	Int32	05/11/12 9:33	05/11/12 9:33	Good
2	B&R X20BC008U	NS2/Numeric1...	DigitalInput	0	Byte	05/11/12 9:33	05/11/12 9:33	Good

Figure 25: Before: DigitalInputs "unpacked"

Figure 26: After: DigitalInputs "packed"

4 OPC UA devices in Automation Studio

It is possible to configure OPC UA devices just like POWERLINK or fieldbus devices in Automation Studio. The advantage of using OPC UA devices is that you don't need device description files for every single device.

With the so-called OPC UA Any Device, it is possible to connect third-party systems to the B&R system. The OPC UA Any Device represents a simple access that can be configured in the Physical View and directly linked to an ETH interface. This device acts as a client and can connect to any server. This means that any OPC UA server – whether it be a controller, a bus controller or an OPC UA-enabled third-party device – can be configured in Automation Studio.

The server information and the variables to be written or read can now be defined directly on this device. These data points are then available via I/O mapping and can be linked to any process variable.

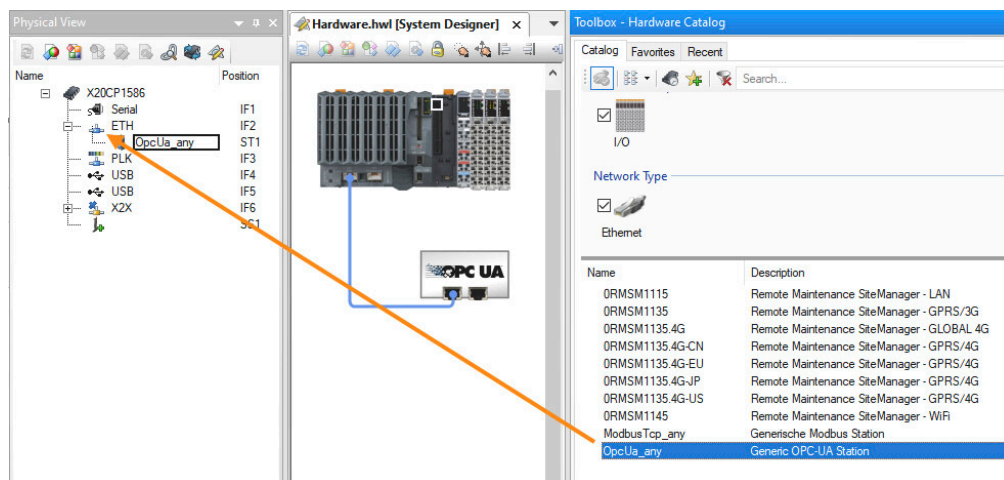


Figure 27: Adding OpcUa_any device from the Toolbox

The OpcUa_any Device, also called generic OPC UA station, is assigned to the Ethernet interface of the CPU in Automation Studio from the Toolbox.



[Communication \ OPC UA \ Configuration in AS \ OPC UA Any Device](#)

All necessary settings are then made in the configuration.

The most important connection parameter is `ServerEndPointUrl`, which is specified with the following syntax:
`opc.tcp://ip-address:port` or `opc.tcp://hostname:port`

Ultimately, this is the same as establishing a connection with UaExpert.

Now it is possible to create channels that are linked to nodes of the OPC UA server. Create a channel by assigning it a name. Then the browse path of the node must be specified with the following syntax: `/namespaceindex:browsername/namespaceindex:browsername/ ...`

The direction (input or output) and the data type must be adapted according to the node that will be used so that the data can be read in or written out correctly.

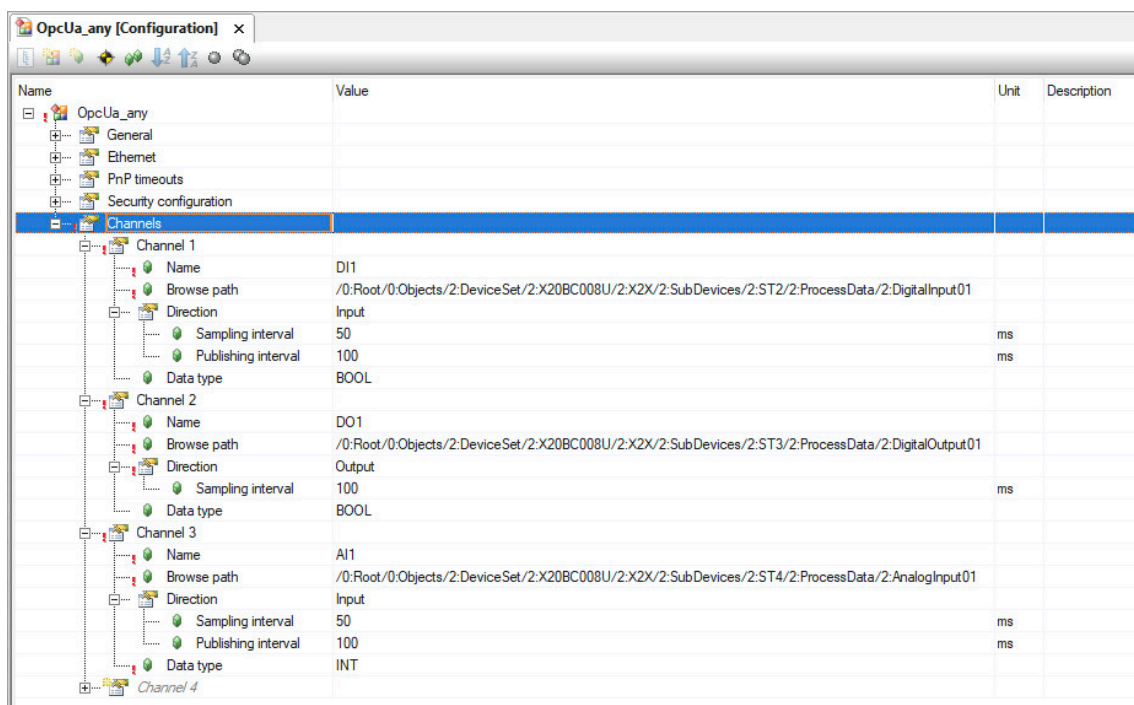


Figure 28: OPC UA station channel settings

After saving the configuration, the previously configured channels appear in the I/O mapping of device OpcUa_any device. Here it is possible to assign process variables to the channels.

Channel Name	Process Variable	Data Type	Task Class	Inverse	Simulate	Source File	Description [1]
+ ModuleOk		BOOL					Module status (1 = module present)
+ ConnectionLostCount		UDINT					Connection errors since last boot
+ ClientErrorCount		UDINT					Error counter for general client errors
+ LastClientError		UDINT					Last client error
+ DI1	::DigitalIn	BOOL	Automatic	☐	☐	\X20CP1586\IoMap i...	
+ DO1	::DigitalOut	BOOL	Automatic	☐	☐	\X20CP1586\IoMap i...	
+ AI1	::AnalogIn	INT	Automatic	☐	☐	\X20CP1586\IoMap i...	

Figure 29: I/O mapping for OPC UA station



Before the project is transferred, it is important to enable the OPC UA system of the CPU. Otherwise, using the generic OPC UA station (OpcUa_any device) will not work!

OPC UA server diagnostics

All events that occur on the OPC UA server are logged in the Logger file "Connectivity". If, for example, the browse path of the channels is entered incorrectly, this will be displayed as a warning in the Logger.

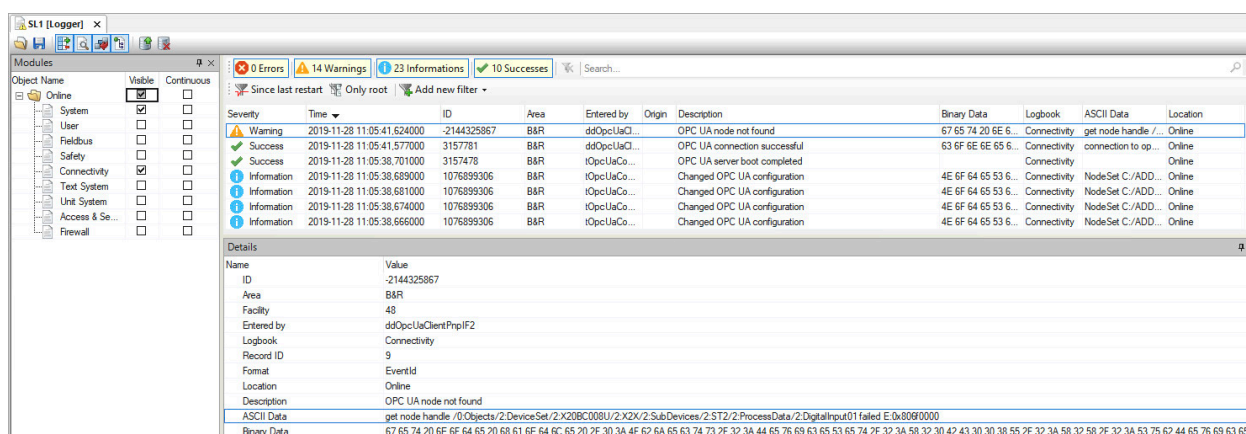


Figure 30: OPC UA server diagnostics: Specified node does not exist

The Logger also logs if the connection to the server was successful.

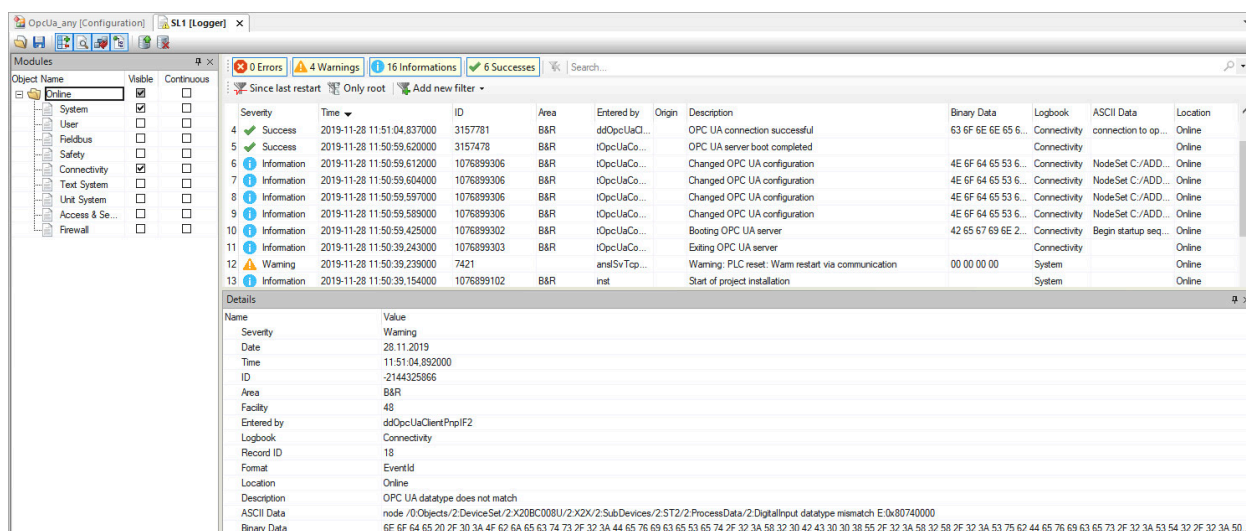


Figure 31: OPC UA server diagnostics: Connection OK



[Diagnostics and service \ Diagnostics tool \ Logger](#)

Exercise: Configure and use OpcUa_any device in a project

This exercise shows how to read variables and values of an OPC UA-capable device at runtime.



The same network topology is still used for this exercise.
It is not necessary to configure the bus controller because of the OpcUa_any device.

- 1) Create a new Automation Studio project with an X20CP1586 controller.
- 2) Configure the Ethernet interface; either static IP address or DHCP.
- 3) Enable the OPC UA system in the CPU configuration, see "Enabling the OPC UA system" on page 23.
- 4) Assign the OpcUa_any device from the Toolbox to the CPU Ethernet interface.
- 5) Enter the ServerEndpointUrl in the OpcUa_any device configuration:
Syntax `opc.tcp://ip-address:port` e.g. `opc.tcp://192.168.1.2:4840`



The default port is 4840. However, this can be changed or read in the CPU configuration (under OPC-UA System \ Activate OPC-UA System \ Network settings).

- 6) Create three channels in the OpcUa_any device configuration, see screenshot above.

Channel 1: /0:Root/0:Objects/2:DeviceSet/2:X20BC008U/2:X2X/2:SubDevices/2:ST2
/2:ProcessData/2:DigitalInput01

Channel 2: /0:Root/0:Objects/2:DeviceSet/2:X20BC008U/2:X2X/2:SubDevices/2:ST3
/2:ProcessData/2:DigitalOutput01

Channel 3: /0:Root/0:Objects/2:DeviceSet/2:X20BC008U/2:X2X/2:SubDevices/2:ST4
/2:ProcessData/2:AnalogInput01



A break in the path is not possible.

- 7) Save OpcUa_any device configuration and check in its I/O mapping whether the channels have been created.
- 8) Transfer the project to the controller.
- 9) Check Logger, see "OPC UA devices in Automation Studio" on page 19.
- 10) Activate monitor mode in the I/O mapping and then observe the value changes of the channels after pressing the switches or turning the potentiometer on the [OPC UA ETA Light](#).
- 11) Force the digital output in the I/O mapping and observe the status change of the LED on the [OPC UA ETA Light](#).



Value changes can be performed and observed in the I/O mapping.

Channel Name	Physical Value	Force Activated	Force Activated Value	Process Variable	Process Variable Value	Data Type
ModuleOk	TRUE	☐	FALSE			BOOL
ConnectionLostCount	0	☐	0			UDINT
ClientErrorCount	1	☐	0			UDINT
LastClientError	2155085824	☐	0			UDINT
DigitalInput01	FALSE	☐	FALSE			BOOL
DigitalOutput01	TRUE	☒	TRUE			BOOL
AnalogInput01	17080	☐	0			INT

Figure 32: Result: I/O mapping

5 Automation Runtime OPC UA server

Enabling the OPC UA system

Automation Runtime offers a completely integrated and easily configurable OPC UA server. This can be activated via an Automation Runtime configuration and the variables can then be released via a separate configuration view. For basic configuration, it's sufficient to enable the OPC UA system. The remaining parameters stay unchanged. The server configuration also offers the option of setting server properties such as the port, encrypted access, transmission cycle times or the number of variables that can be linked. The OPC UA system must be activated for operation as a server as well as a client.

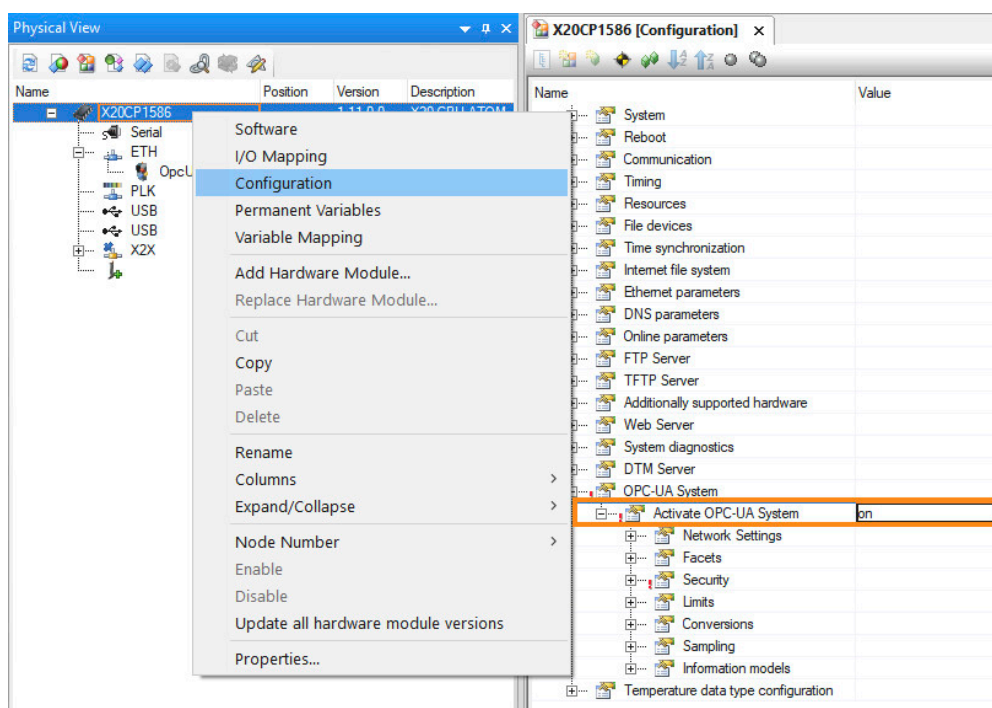


Figure 33: Configuration of the OPC UA system



[Communication \ OPC UA \ Configuration in AS \ Enable OPC UA system](#)

Access & Security - User management system

The Configuration View contains folder "Access & Security/UserRoleSystem" in which two configuration elements are stored.

Basic roles for the system can be created in the "Role" file. These can be an operator, a service technician or an administrator, for example.

No new roles can be created at runtime.

In the "User" file, individual users can then be created to which one or more roles can be assigned.

Users as well as the associated roles can be adapted at runtime.



Do not delete or rename existing roles "Administrators" and "Everyone".
Also do not delete or rename existing users "Admin" and "Anonymous".



[Programming \ Access & Security \ User role system \ Configuration \ Automation Studio configuration](#)

Exercise: Configure the Automation Runtime OPC UA server and create users.

The objective of this exercise is to configure a user role system for the OPC UA server. The roles "ServiceEngineer" and "Admin" and a user "Mike" are created.

- 1) Enable the OPC UA system in the CPU configuration.
- 2) Create role "Service engineer" in the Configuration View under AccessAndSecurity\UserRoleSystem.
- 3) Create user "Admin" with role "Admin" and password "Admin".
- 4) Create user "Mike" with role "ServiceEngineer" and password "Mike".



The configuration of the user role system should correspond to the following figure:

The screenshot displays two side-by-side configuration windows from the OPC UA software.

Left Window: Role.role [Role Configuration]

- Current 'role' element: No element selected.
- Tree view shows three roles:
 - Administrators** (Role ID: 1, Description: Role for..)
 - Everyone** (Role ID: 2, Description: Role for..)
 - ServiceEngineer** (Role ID: 3, Description: Role for..)

Right Window: User.user [User Configuration]

- Current 'user' element: No element selected.
- Tree view shows three users:
 - Anonymous** (User ID: 1, Password: , Roles: Assigned Role [1] Everyone, Assigned Role [2])
 - Admin** (User ID: 2, Password: *****, Roles: Assigned Role [1] Administrators, Assigned Role [2])
 - Mike** (User ID: 3, Password: *****, Roles: Assigned Role [1] ServiceEngineer, Assigned Role [2])

Figure 34: Result: User role system

5.1 Configuration of the default view

Adding OPC UA mapping

The OPC UA default view will be inserted from the Toolbox into the Connectivity folder of the Configuration View. In this view, all variables of the corresponding configuration are displayed for approval. This means that the Automation Runtime OPC UA server can then make these available.

In the OPC UA default view, many settings can also be made for ranges of values, write and read permissions, units and much more.

The OPC UA default view is added in the Configuration View under Connectivity/OpcUa from the Toolbox.

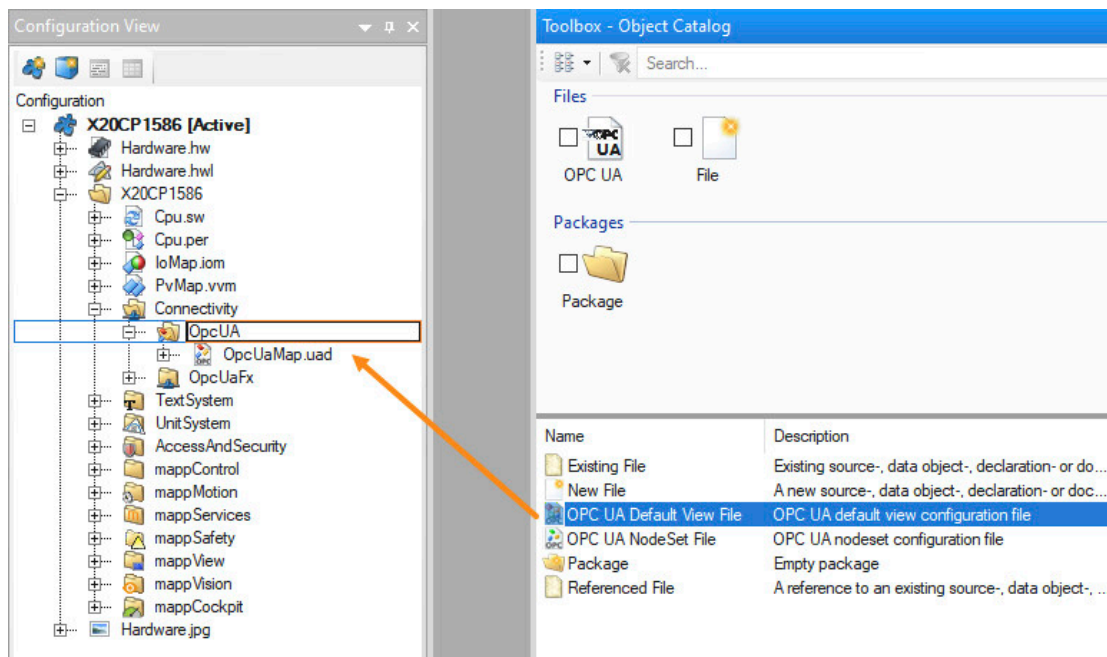


Figure 35: Adding the OPC UA default view from the Toolbox

Releasing variables

After opening the OPC UA default view, the global and local process variables, which should be visible as OPC UA tags on the client, are enabled. As soon as the variables are enabled, the access rights, the limits, the activation of the history and the unit can be defined via the Properties window.

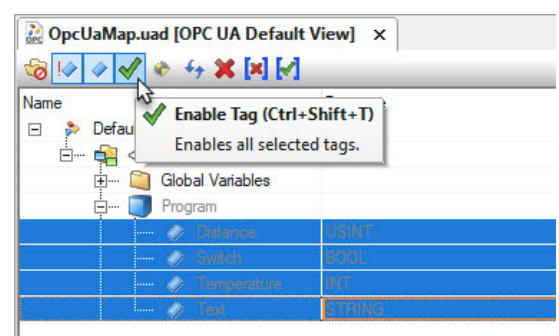


Figure 36: OPC UA default view - Enabling a tag

Link to user administration

The access restrictions to the OPC UA system are based on the user management integrated in Automation Runtime and can be configured directly in the OPC UA default view.

To do this, the roles authorized for the overall system must first be specified on level "Default View". The permissions that are assigned for the individual roles are inherited to all released data points, but can still be overwritten on the properties of all data points.

After clicking on "Default View", the desired roles are added in the Properties window via "Add role".

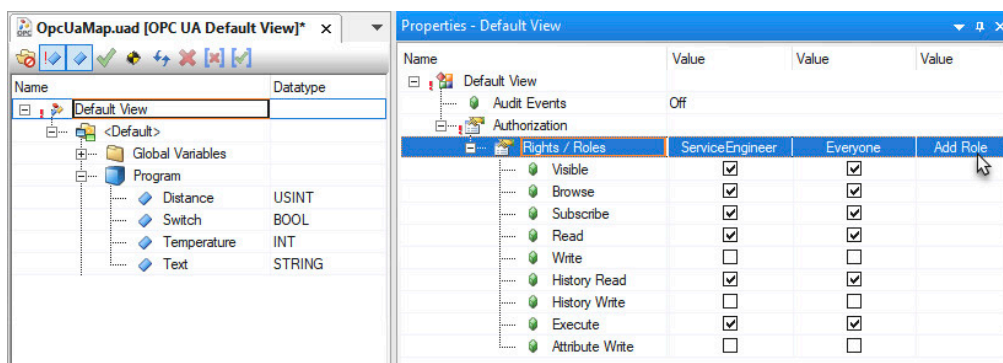


Figure 37: OPC UA default view - Adding roles



[Communication / OPC UA / Configuration in AS / OPC UA default view configuration / OPC UA default view configuration / Properties of OPC UA tags / Authorization Programming \ Access & Security](#)

Exercise: Create a program and initialize variables

A new Structured Text program is created and the following variables are added: Temperature (INT), Distance (USINT), Switch (BOOL) and Text (STRING).

In the initialization subroutine, the variables are set to any values.

In the cyclic program, the distance variable is increased by one if the switch is set to TRUE.

- 1) Add a new ST program.
- 2) Declare variables.
- 3) Initialize variables in the initialization subroutine.
- 4) Program the logic; cyclically increase variable "Distance" by one if the switch is active.
- 5) Transfer program.



The logic should be programmed and the variables created as follows.

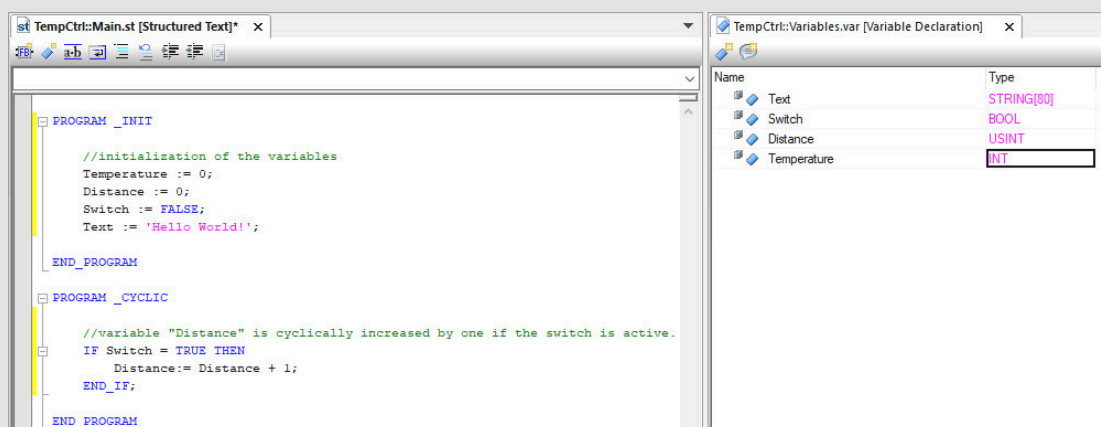


Figure 38: Result: Creating a program

Exercise: Enable the OPC UA default view

The OPC UA default view is added to the configuration. This allows the variables created in the program to be released on the Automation Runtime OPC UA server.

- 1) Add the OPC UA default view in the Configuration View under Connectivity\OpcUA using the Toolbox.
- 2) Select the default view and add all roles in the properties window.

- 3) Configure write permissions for the Administrator and ServiceEngineer roles.
- 4) Enable variables "Temperature", "Distance", "Switch" and "Text", see "Releasing variables" on page 25.

5.2 Applying units, users, limits and historizing

Assigning units

After selecting a variable in the OPC UA default view, a unit can be assigned to the variable via the Toolbox. For this there is a large number of predefined units that can be searched via the search bar or filter settings.

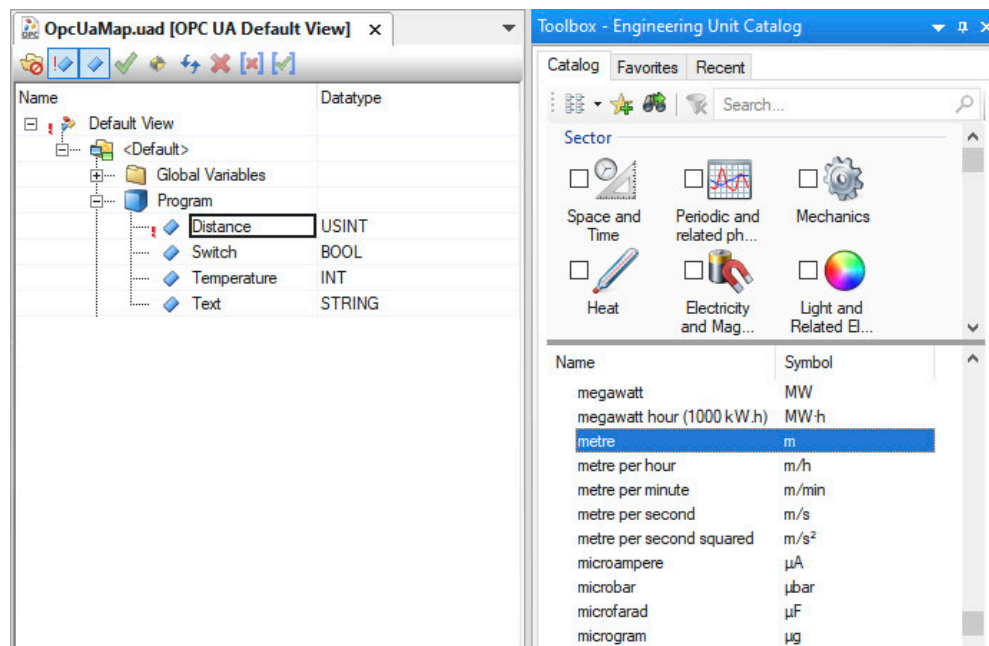


Figure 39: Toolbox - Assigning units

Limits

Limit values can also be assigned to the nodes. Generally valid default limit values can be assigned, or also role-related ones. The desired roles can be added via "Add role". A role-related limit value can then be defined.



[Communication / OPC UA / Configuration in AS / OPC UA default view configuration / OPC UA default view configuration / Properties of OPC UA tags / Range of values / Role-based range of values](#)

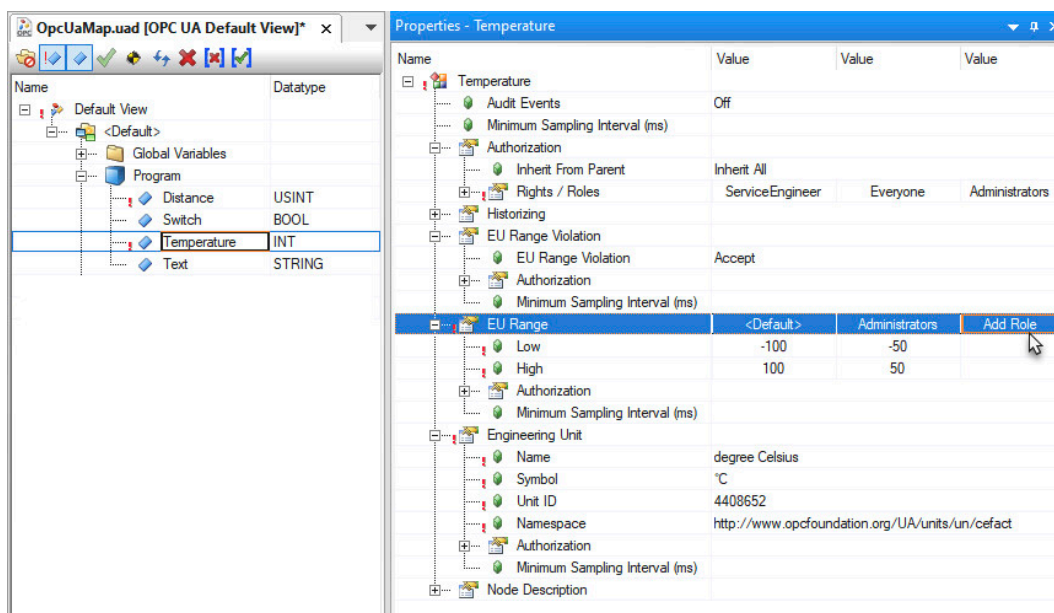


Figure 40: Properties - Adding limits

There are different ways of handling value range violations. They can be accepted (Accept), rejected (Reject) or clamped (Clamp). In the case of the latter, the maximum or minimum value is applied if the value range is violated.

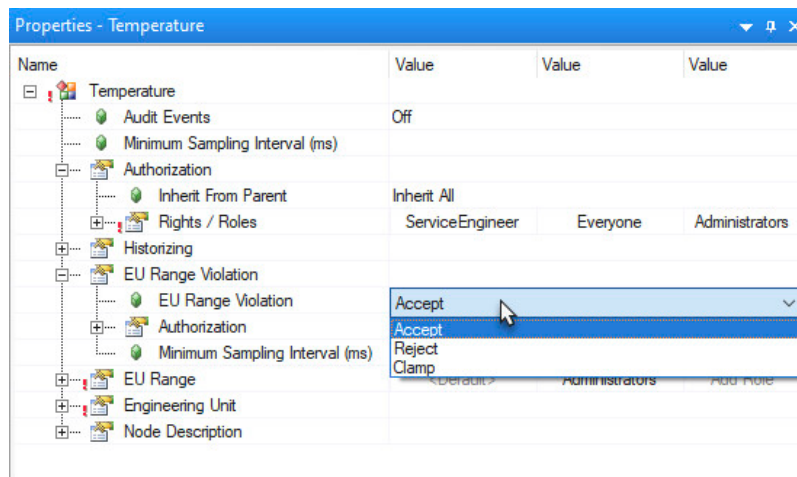


Figure 41: Properties - Accepting/Rejecting value range violations

Rights and roles

Each node can be assigned its own role and rights. For example, nodes can only be written by administrators. Either the rights/roles of the previous instance can be inherited, or you can set them yourself.

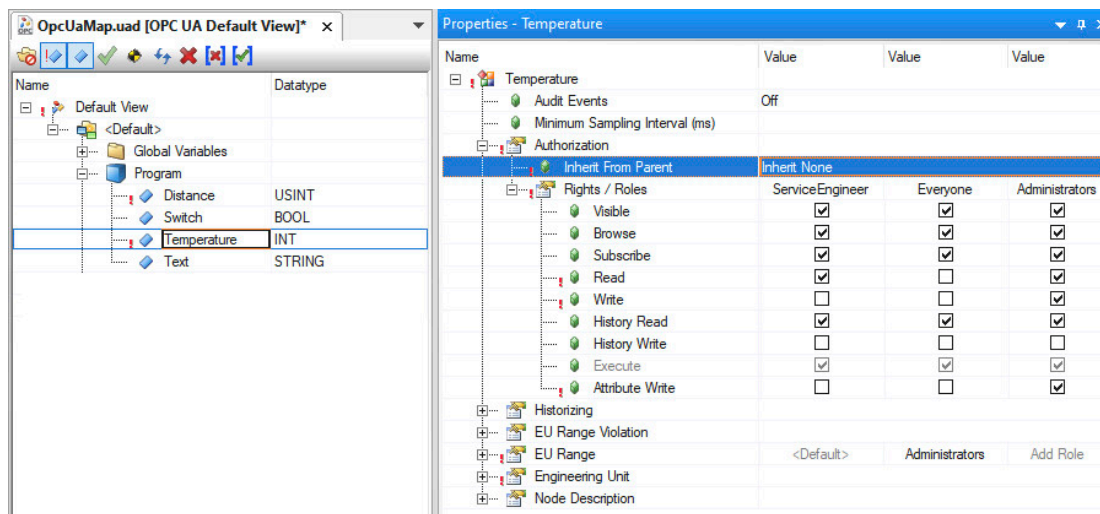


Figure 42: Properties - Rights/Roles

Historizing

It is possible to enable the historizing function for each node. If this is enabled, the values are written to a buffer at a certain time interval. This buffer can then be read and the values of the node can be analyzed.

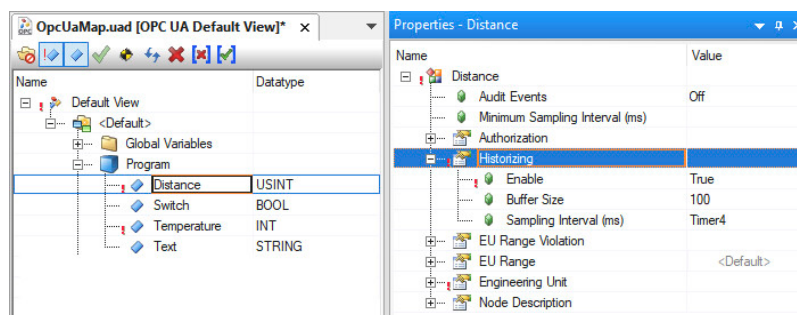


Figure 43: Properties - Historizing

Exercise: Configure units, limits and rights

The OPC UA default view has already been added to the configuration. Now, further properties of the OPC UA tags are configured.

- 1) Assign a unit to variables "Distance" and "Temperature".
- 2) Assign limits to variables "Distance" and "Temperature".
- 3) Add role-dependent settings to variables "Distance" and "Temperature" for the value range.
- 4) Transfer the project.

The following attributes are assigned to variable "Distance":

- Unit: "meter"
- Default value range: 0 / 10
- Value range violation: Reject

The following attributes are assigned to variable "Temperature":

- Unit: "degree Celsius"
- Default value range: -100 / 100
- "Administrator" value range: -50 / 50
- "Service engineer" value range: -20 / 20
- Default value range "Everyone": -10 / 10
- Value range violation: Truncate



The properties of the variables should correspond to the following figures.

Properties - Distance			
Name	Value	Value	Value
Distance			
Audit Events	Off		
Minimum Sampling Interval (ms)			
Authorization			
Historizing			
EU Range Violation			
EU Range Violation	Reject		
Authorization			
Minimum Sampling Interval (ms)			
EU Range	<Default>	Add Role	
Low	0		
High	10		
Authorization			
Minimum Sampling Interval (ms)			
Engineering Unit			
Name	metre		
Symbol	m		
Unit ID	5067858		
Namespace	http://www.opcfoundation.org/UA/units/un/c...		
Authorization			
Minimum Sampling Interval (ms)			
Node Description			

Figure 44: Properties of "Distance"

Properties - Temperature				
Name	Value	Value	Value	Value
Temperature				
Audit Events	Off			
Minimum Sampling Interval (ms)				
Authorization				
Historizing				
EU Range Violation				
EU Range Violation	Clamp			
Authorization				
Minimum Sampling Interval (ms)				
EU Range	<Default>	Administrators	ServiceEngineer	Everyone
Low	-100	-50	-20	-10
High	100	50	20	10
Authorization				
Minimum Sampling Interval (ms)				
Engineering Unit				
Name	degree Celsius			
Symbol	°C			
Unit ID	4408652			
Namespace	http://www.opcfoundation.org/UA/units/un/cefact			
Authorization				
Minimum Sampling Interval (ms)				
Node Description				

Figure 45: Properties of "Temperature"

Exercise: Use UaExpert to test the limits of the variables with different users

A connection is established with UaExpert to the Automation Runtime OPC UA server. The set limits are checked by logging in via different users.

- 1) Establish a connection to the Automation Runtime OPC UA server.
- 2) Navigate to the following path in the address space: Root/Objects/PLC/Modules/<Default>/Program.
- 3) Add the variables to "Data Access View".
- 4) Change the values of the variables with different users and observe their behavior.



The user can be changed either in the shortcut menu of the server or at the top of the toolbar. User-name and password depend on the configured roles in the OPC UA default view in the Automation Studio project.

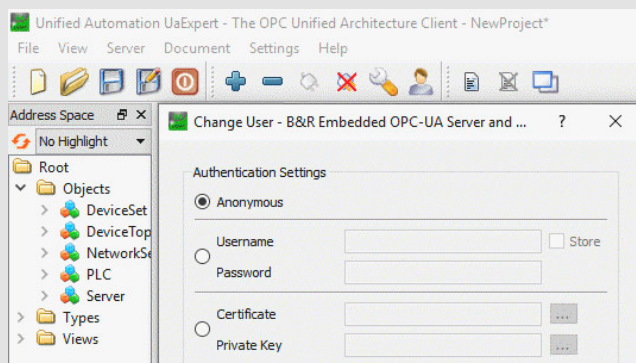


Figure 46: UaExpert: Changing the user



If user "Anonymous" with role "Everyone" wants to enable variable "Switch", this is prevented since this user does not have the respective rights.

Data Access View									
#	Server	Node Id	Display Name	Value	Datatype	Source Timestamp	Server Timestamp	Statuscode	
1	B&R Embedded...	NS6[String]::Pro...	Distance	31	Byte	10:19:05.414	10:19:05.414	Good	
2	B&R Embedded...	NS6[String]::Pro...	Switch	false	Boolean	10:19:05.414	10:19:05.414	Good	
3	B&R Embedded...	NS6[String]::Pro...	Temperature	20	Int16	10:18:49.649	10:18:49.649	Good	
4	B&R Embedded...	NS6[String]::Pro...	Text	Hello World	String	10:18:49.649	10:18:49.649	Good	

Log			
Timestamp	Source	Server	Message
04.06.2019 08:19:21.588	DA Plugin	B&R Embedded OPC-UA Server and Client	Write to node 'NS6[String]::Program:Switch' failed [ret = BadUserAccessDenied]

Figure 47: Result: Testing variables

Exercise: Historizing data points

In this exercise, historizing variable "Distance" is enabled. This makes it possible to view the history of a variable over time under "History Trend View" in UaExpert.

- 1) Select variable "Distance" and open the properties window.
- 2) Enable historizing.
- 3) Set buffer size to 100.
- 4) Set sampling interval to timer4.
- 5) Transfer the project.
- 6) Update the configuration in UaExpert (right click on "Root - Rebrowse").
- 7) Enable variable "Switch" so that variable "Distance" is increased cyclically.
- 8) Add "History Trend View".

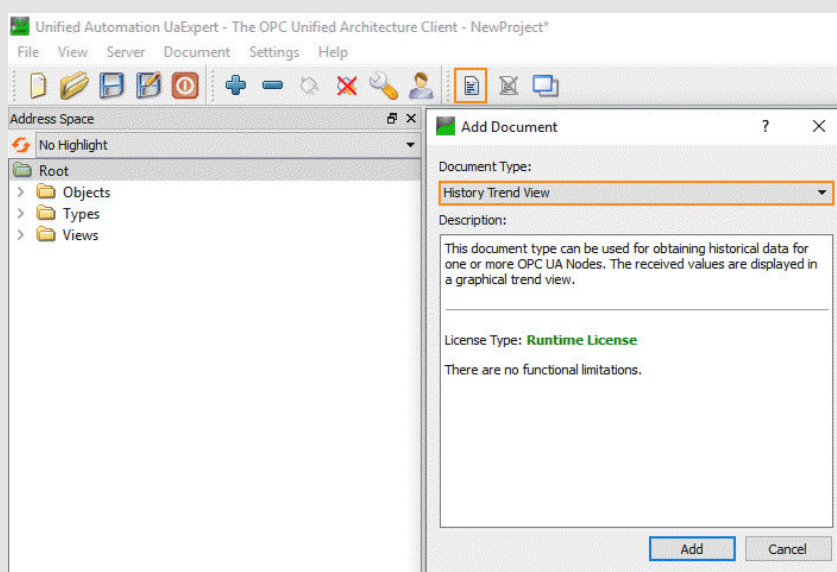


Figure 48: UaExpert: Adding "History Trend View"

- 9) Add variable "Distance" to "History Trend View" via drag-and-drop.
- 10) Start the cyclic update.



You can now view the characteristic curve over time of a variable in "History Trend View". Because variable "Distance" is of data type USINT, the value is set to 0 if more than 255 is counted.

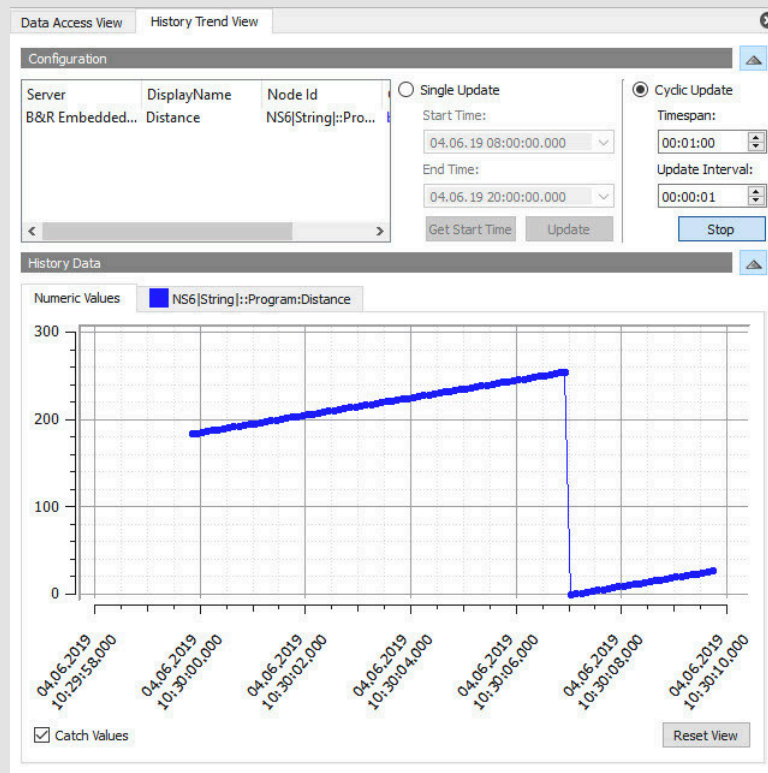


Figure 49: Result: Historizing

6 Certificates and security

Certificates

Instead of renewing the certificates, the OPC Foundation utilizes the functionality of TLS¹ Layers For additional information, see Automation Help.

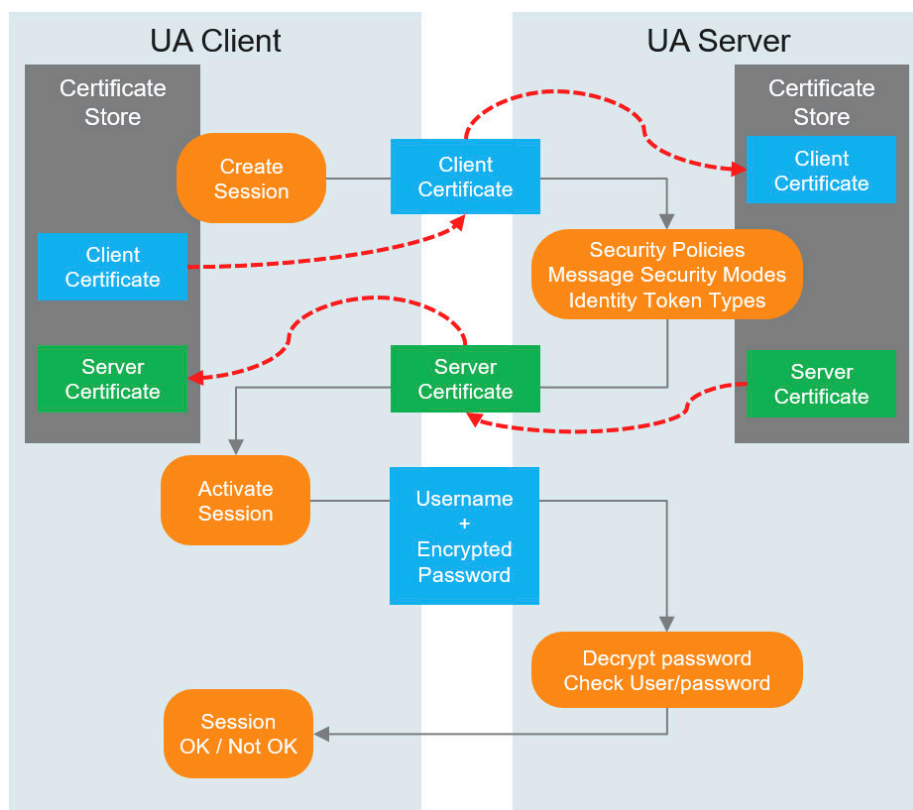


Figure 50: Handling certificates



[Programming / Access & Security / Transport Layer Security \(TLS\)](#)

¹ Transport Layer Security (TLS, more widely known under the predecessor name Secure Sockets Layer (SSL), is a hybrid encryption protocol for secure data transmission on the Internet (en.wikipedia.org/wiki/Transport_Layer_Security).

To create a certificate in Automation Studio, a new certificate is added to the Configuration View in the AccessAndSecurity \ CertificateStore \ OwnCertificates \ Certificates package from the Toolbox.

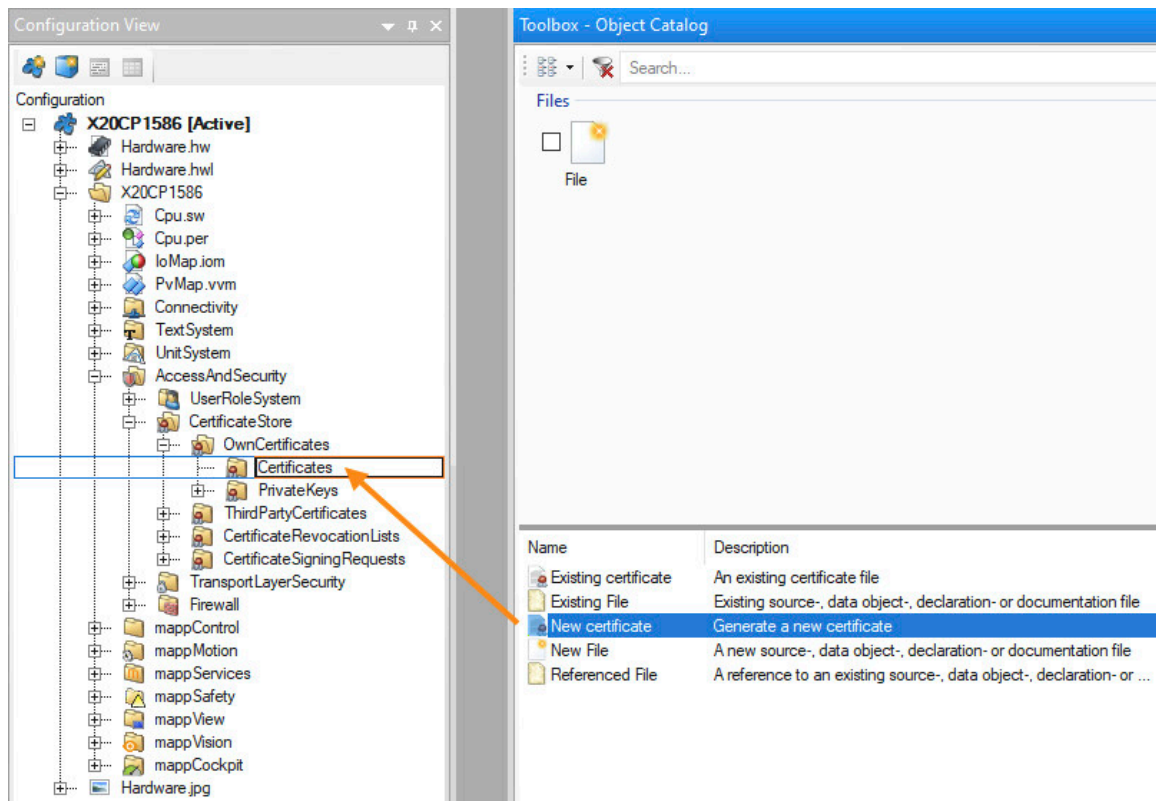


Figure 51: Pasting the certificate

When generating the certificate, the data can now be entered with the corresponding requirements.

The 'Add OPC UA Certificate' dialog box is shown. It contains the following fields and options:

- Key size:** Radio buttons for 1024 bit (selected), 2048 bit, and 4096 bit.
- 2-Digit country code*:** Text field with 'AT'.
- Password:** Text field with 'password'.
- Expiry date*:** Date picker showing '17.07.2021', '2' years, and '0' days.
- Full state or province:** Text field with 'Austria'.
- Locality (City):** Text field with 'Eggelsberg'.
- Organization:** Text field with 'Academy'.
- Organization unit:** Text field.
- Contact E-Mail:** Text field with 'contact@email.com'.
- Common name (Server address):** Text field with 'Academy Server'.
- URI:** Text field with 'urn:br-automation/BR-UA/EmbeddedServer' and a 'Default' button.
- DNS:** Radio button selected, text field with 'br-automation'.
- IP:** Radio button unselected, text field with '192.168.1.98'.

At the bottom, there is a note: '(*) Mandatory fields' and 'Generating the certificate may take some time depending on the key size.' The 'Finish' button is highlighted.

Figure 52: Configuring the certificate

In order to use the certificate, it must be assigned to an SSL configuration. This is added in the Configuration View in the package AccessAndSecurity/TransportLayerSecurity from the Toolbox.

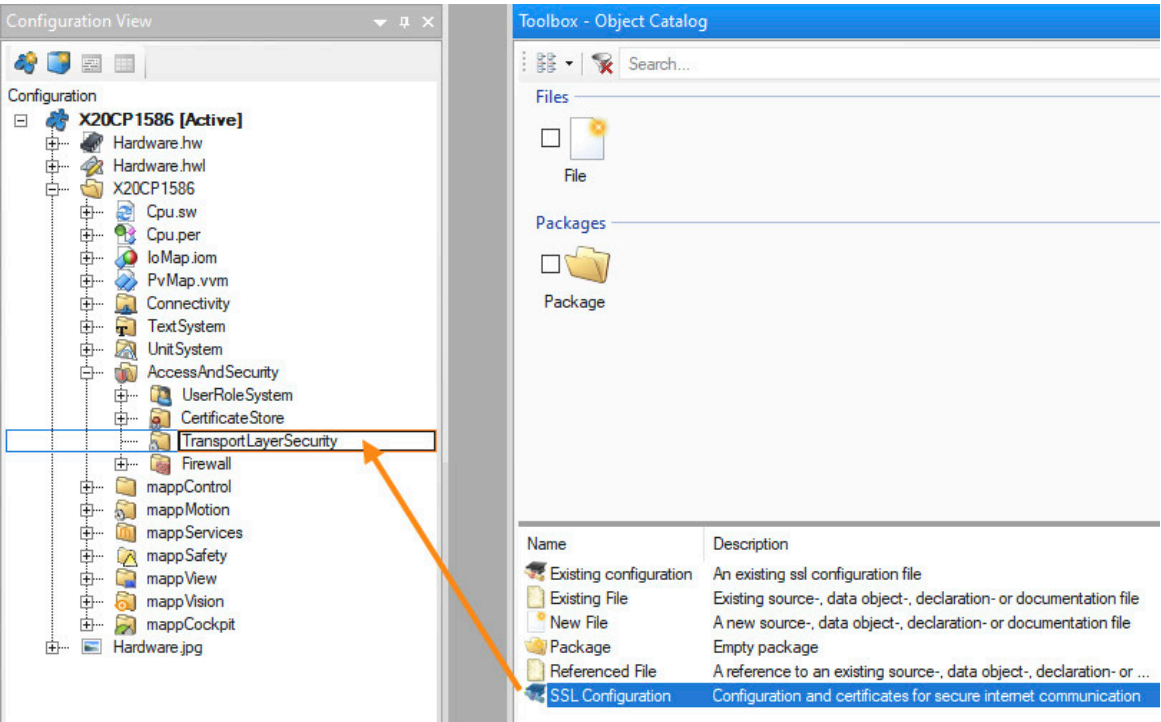


Figure 53: Adding the SSL configuration

In this SSL configuration, the configuration type must now be changed from "General SSL configuration" to "OPC UA SSL configuration". Then the certificate, the key and the password of the certificate are assigned.

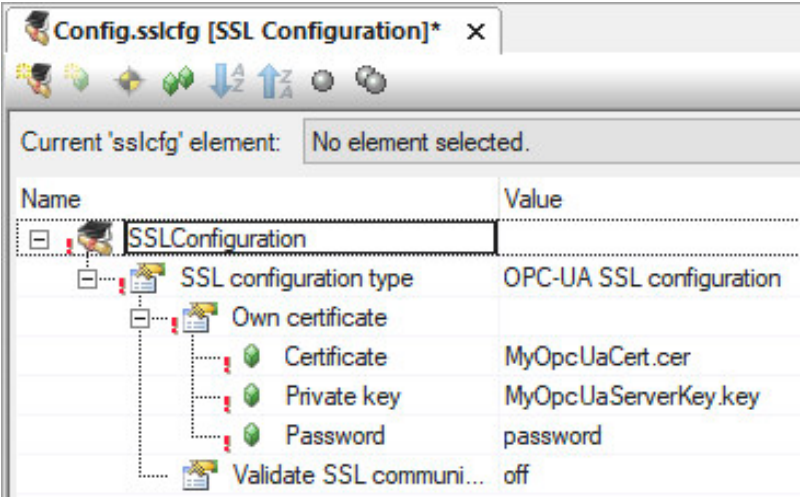


Figure 54: Configuring the SSL configuration

The SSL configuration must be assigned to the OPC UA server so that the Automation Runtime OPC UA server of the controller can then take over the certificate.

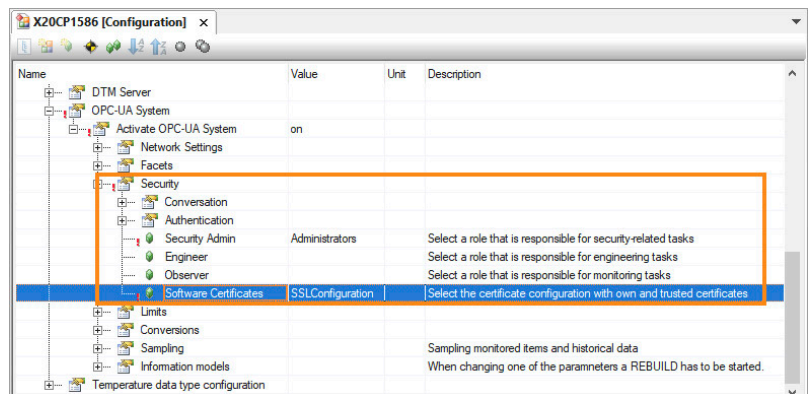


Figure 55: Assigning the SSL configuration

Exercise: Create and use your own certificates

In this exercise, you will create your own certificate and test it with UaExpert.

- 1) Under AccessAndSecurity \ CertificateStore \ OwnCertificates \ Certificates, add a new OPC UA certificate from the Toolbox.
- 2) Enter data and complete the certificate creation.
- 3) Add an SSL configuration under AccessAndSecurity\TransportLayerSecurity from the Toolbox.
- 4) In field "SSL configuration type", select option "OPC UA SSL configuration" and assign the created certificate, private key and password.
- 5) In the CPU configuration under OPC UA System \ Activate OPC UA System \ Security, select option "SSLConfiguration" in field "Software Certificates".
- 6) Transfer the project and use UaExpert to confirm the new certificate when establishing the connection.



When establishing a connection with the UaExpert client, the user's own certificate can be confirmed.

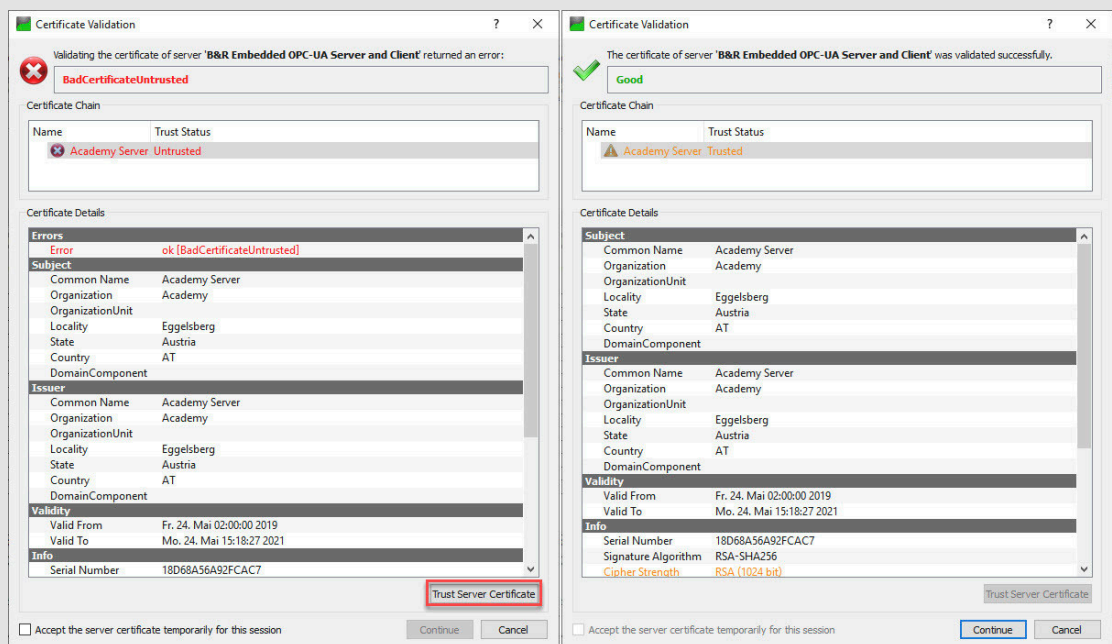


Figure 56: Confirming the certificate

6.1 Trusted list

A trust list can be stored on the Automation Runtime OPC UA server. First, the certificates you would like to specify as trusted are added to the package AccessAndSecurity \ CertificateStore \ Third Party Certificates \ SoftwareCertificates from the Toolbox.

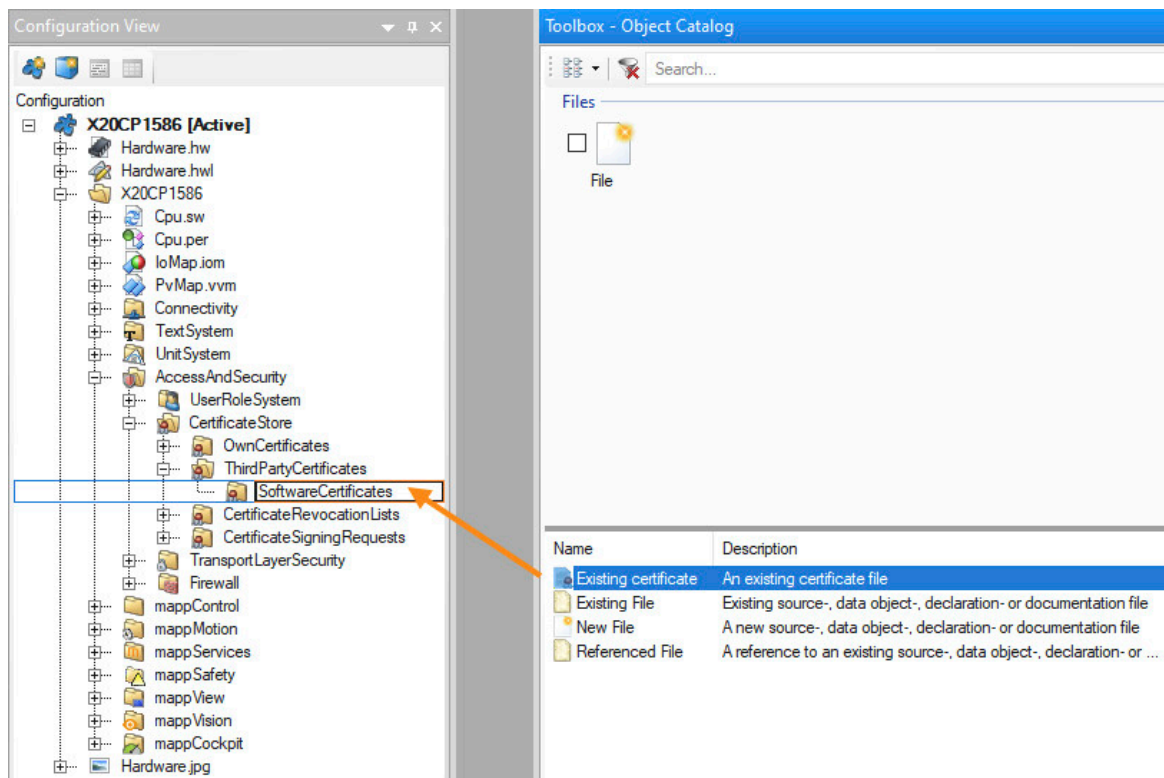


Figure 57: Third-party certificates

Next, item "Validate SSL communication partners" is enabled and the desired certificates are assigned. This corresponds to the trusted list. The OPC UA server with this SSL configuration only trusts clients that have a certificate from this trusted list.

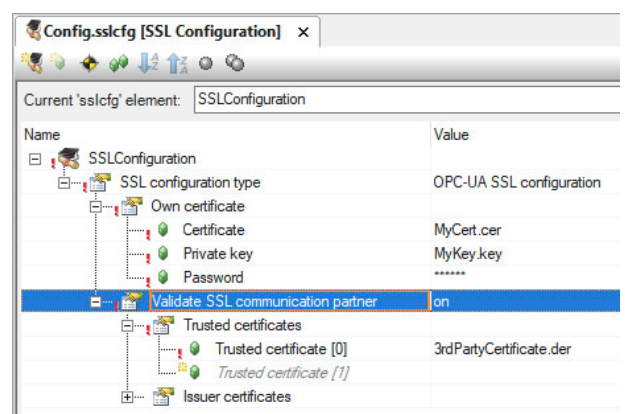


Figure 58: Trusted list

Exercise: Attempt to establish an encrypted connection without a client certificate on the server

In this exercise, attempt to establish an encrypted connection without a client certificate on the server.

- 1) Enable data point "Validate SSL communication" in the SSL configuration under AccessAndSecurity \ Transport-LayerSecurity.
- 2) Adjust the safety guidelines in the configuration of the OPC UA system as shown.

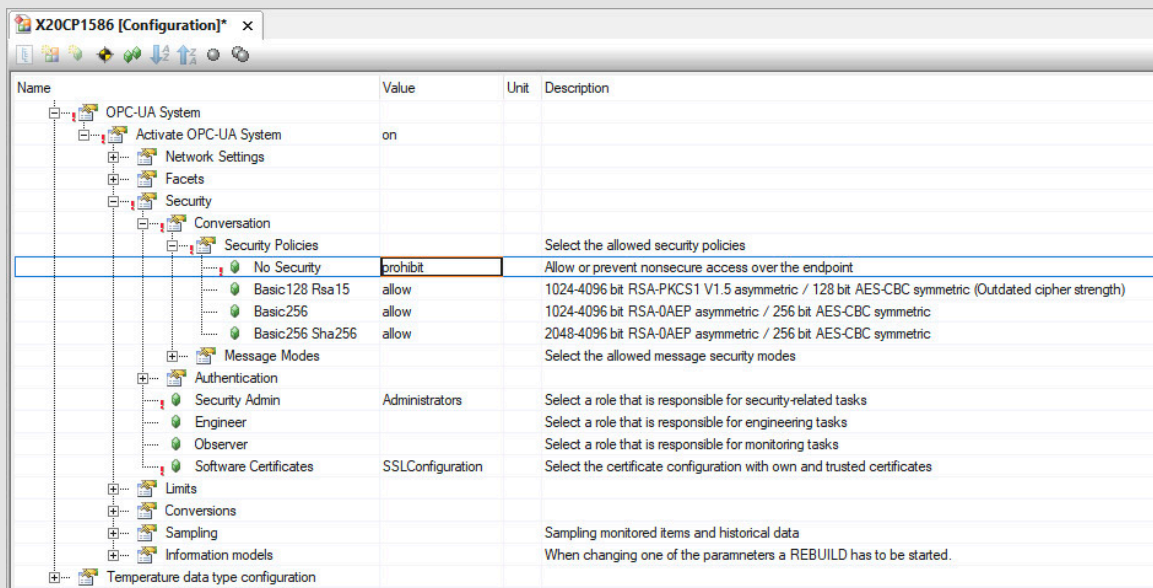


Figure 59: Safety guidelines

- 3) Transfer the project.
- 4) Connect to the server.

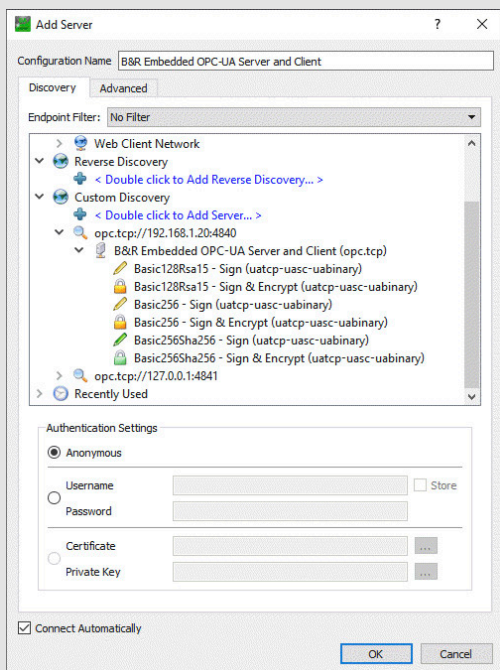


Figure 60: Establishing an encrypted connection

- 5) Check if the connection is rejected.



If the UaExpert certificate is not in the server's trusted list, no connection can be established between the UaExpert client and the OPC UA server.

Log			
Timestamp	Source	Server	Message
04.06.2019 09:33:11.419	Server Node	B&R Embedded OPC-UA Server and Client	Endpoint: 'opc.tcp://br-automation:4840'
04.06.2019 09:33:11.419	Server Node	B&R Embedded OPC-UA Server and Client	Security policy: 'http://opcfoundation.org/UA/SecurityPolicy#Basic256'
04.06.2019 09:33:11.419	Server Node	B&R Embedded OPC-UA Server and Client	ApplicationUri: 'urn:br-automation/BR/UA/EmbeddedServer'
04.06.2019 09:33:11.419	Server Node	B&R Embedded OPC-UA Server and Client	Used UserTokenType: Anonymous
04.06.2019 09:33:11.465	Server Node	B&R Embedded OPC-UA Server and Client	Error 'BadSecurityChecksFailed' was returned during OpenSecureChannel
04.06.2019 09:33:11.465	Server Node	B&R Embedded OPC-UA Server and Client	Connection status of server 'B&R Embedded OPC-UA Server and Client' changed to 'Disconnected'.

Figure 61: Result: Trusted list

Exercise: Use the trusted list

The objective of this exercise is to use the trusted list. For this, the UaExpert certificate is added to the trusted list. The final step is a test to see whether it is possible to establish an encrypted connection.

- 1) Select "Own certificate" in UaExpert under Settings\Manage Certificates and save it in the project folder via command "Copy application certificate to".

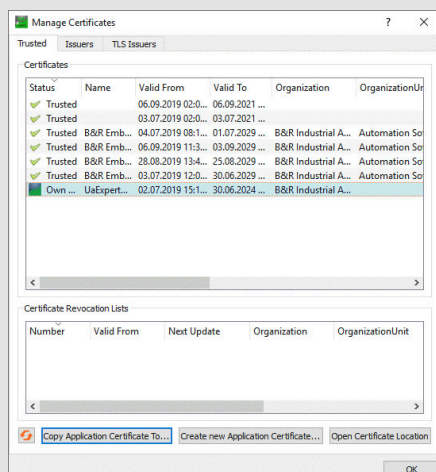


Figure 62: UaExpert: Managing certificates

- 2) Add the saved UaExpert certificate in the Configuration View under AccessAndSecurity \ CertificateStore \ ThirdPartyCertificates \ SoftwareCertificates.
- 3) Add the UaExpert certificate to the trust list in the SSL configuration.
- 4) Transfer the project.
- 5) Establish an encrypted connection using UaExpert.
- 6) Check if the connection is accepted.



If the client certificate (from UaExpert) was added to the trusted list and attempts to establish a connection, this can now be done successfully.

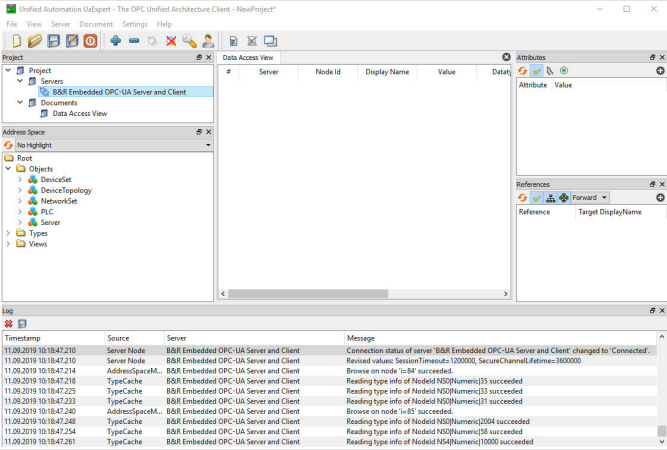


Figure 63: Connection setup successful

7 Publisher/Subscriber configuration

Publish-subscribe model

OPC UA used to work with a client/server mechanism, where a client sends a request to the server. The server processes the request and then sends a response back to the client. The client/server system will reach its limits if there are many too nodes on the network.

That was the reason for adding a publish-subscribe architecture to the OPC UA communication solution. The publish-subscribe model enables one-to-many and many-to-many communication. A server sends its data to the network (publish) and every client can receive this data (subscribe).

With a client-server mechanism, a client requests information and receives a response from a server.

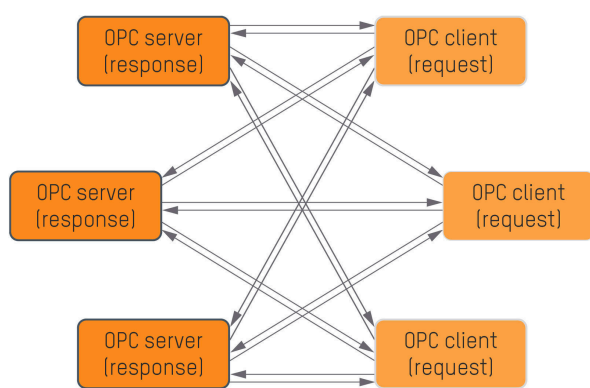


Figure 64: Client/Server system

With a publish-subscribe model, a server sends its data to the network (publish) and every client can receive this data (subscribe).

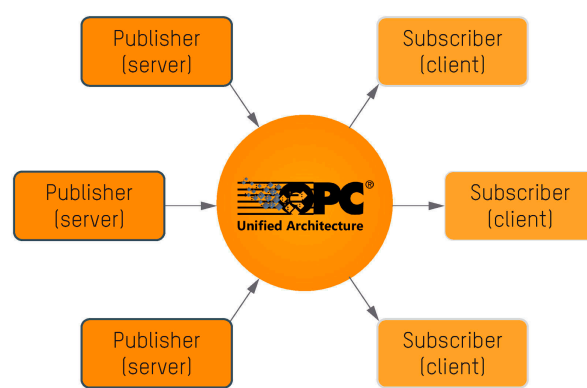


Figure 65: Publish-subscribe model

Publisher configuration

For information about how to configure the Publisher, see Automation Help.



[Communication \ UPC UA FX \ PubSub \ PubSub configuration in Automation Studio \ PubSub publisher configuration](#)

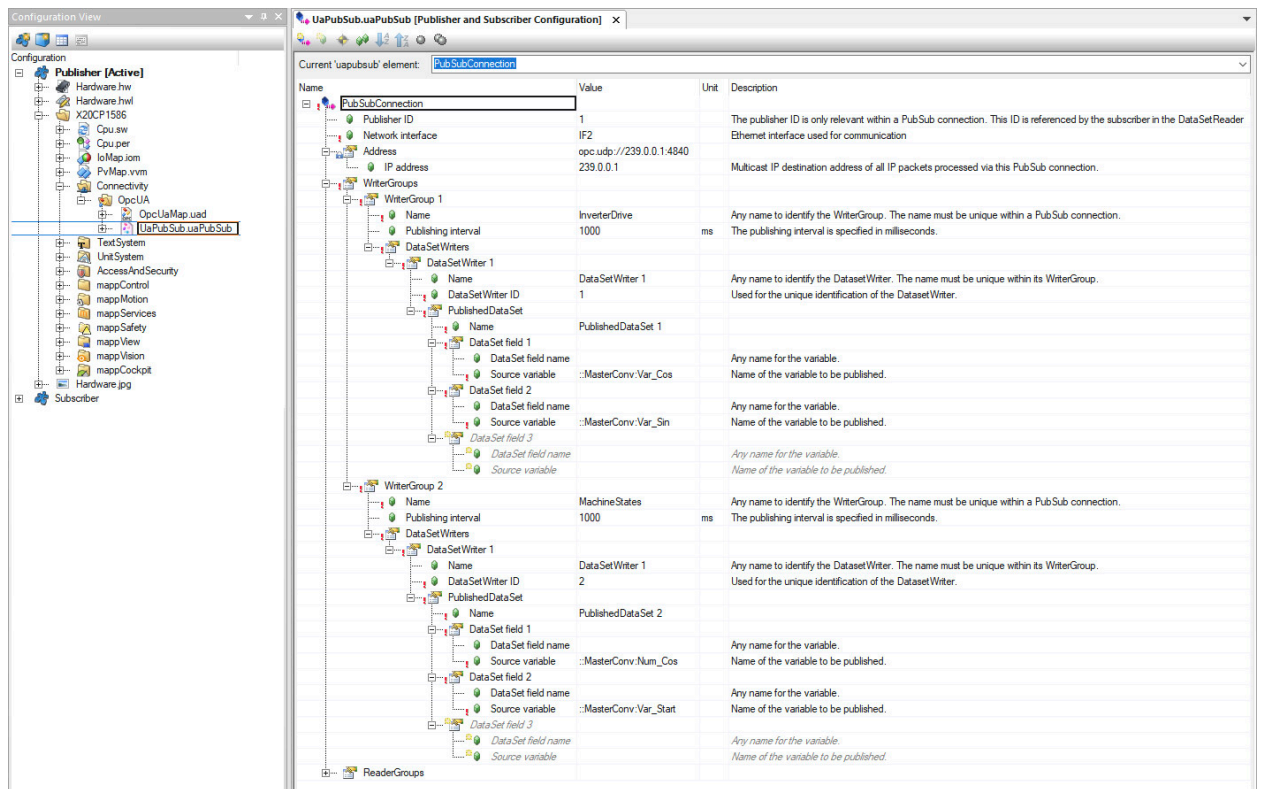


Figure 66: Publisher configuration

The following program is used in the publisher configuration.

Source code: MasterConveyor

Variables: MasterConveyor

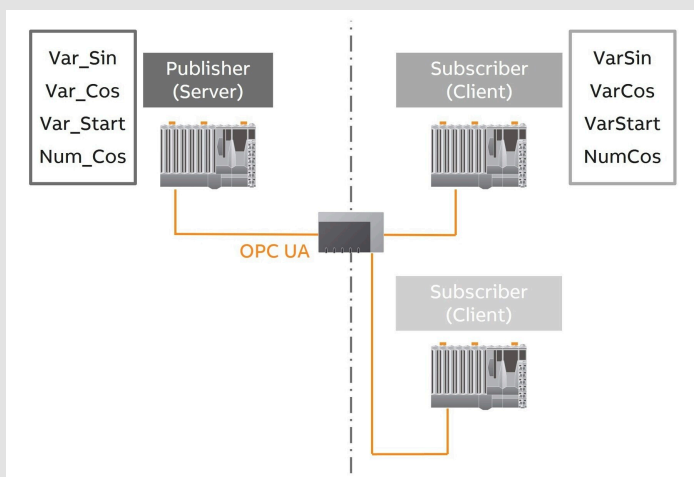
```
PROGRAM _CYCLIC
(*Standard application
- counting number /
calculate sin/cos*)
Var_Start := Var_Start+0.1;
Var_Sin := SIN(Var_Start);
Var_Cos := COS(Var_Start);
IF Var_Cos > 0.999 THEN
    Num_Cos := Num_Cos+1;
END_IF;
END_PROGRAM
```

```
VAR
    Var_Cos : REAL;
    Var_Sin : REAL;
    Num_Cos : REAL;
    Var_Start : REAL;
END_VAR
```

Exercise: Configure a subscriber

Statistical data accumulates on a control system. This data is needed on several machines. In this exercise, participants will learn how to send data back and forth between different devices using a Pub/Sub model.

- 1) Connect controllers to a DHCP network.



- 2) Create a new ST program and variables for SlaveConveyor.

```
VAR
    VarStart : REAL;
    VarSin : REAL;
    VarCos : REAL;
    VarNum : REAL;
END_VAR
```

- 3) Use variables in the init program so they are created by the compiler.
- 4) Configure subscriber, see Automation Help.



[Communication \ UPC UA FX \ PubSub \ PubSub configuration in Automation Studio \ PubSub subscriber configuration](#)

- 5) Transfer configurations and test communication.



Name	Value	Unit	Description
PubSubConnection			
Publisher ID	1		The publisher ID is only relevant within a PubSub connection. This ID is referenced by the subscriber in the DataSetReader Ethernet interface used for communication
Network interface	IF2		
Address	opc.tcp://239.0.0.1:4840		
IP address	239.0.0.1		Multicast IP destination address of all IP packets processed via this PubSub connection.
ReaderGroups			
ReaderGroup 1			
Name	ReaderGroup 1		Any name to identify the reader group. The name must be unique within a PubSub connection.
DataSetReaders			
DataSetReader 1			
Name	DataSetReader 1		Any name to identify the DataSetReader. The name must be unique within its reader group.
SubscribedDataSet			
Name	PubSubConnection-InverterDrive-DataSetWriter 1		Any name to identify the dataset. The name must be unique within its connection.
DataSet field 1			
DataSet field name	MasterConv-Var_Cos		Name of the subscribed variable (name on the publisher page).
Target variable	:SlaveConv-VarCos		Name of the OPC UA tag to which the subscribed variable is mapped.
DataSet field 2			
DataSet field name	MasterConv-Var_Sin		Name of the subscribed variable (name on the publisher page).
Target variable	:SlaveConv-VarSin		Name of the OPC UA tag to which the subscribed variable is mapped.
ReaderGroup 2			
Name	ReaderGroup 2		Any name to identify the reader group. The name must be unique within a PubSub connection.
DataSetReaders			
DataSetReader 1			
Name	DataSetReader 1		Any name to identify the DataSetReader. The name must be unique within its reader group.
SubscribedDataSet			
Name	PubSubConnection-MachineStates-DataSetWriter 1		Any name to identify the dataset. The name must be unique within its connection.
DataSet field 1			
DataSet field name	MasterConv-Num_Cos		Name of the subscribed variable (name on the publisher page).
Target variable	:SlaveConv-VarNum		Name of the OPC UA tag to which the subscribed variable is mapped.
DataSet field 2			
DataSet field name	MasterConv-Var_Start		Name of the subscribed variable (name on the publisher page).
Target variable	:SlaveConv-VarStart		Name of the OPC UA tag to which the subscribed variable is mapped.

Figure 67: Subscriber configuration

8 PLCopen function blocks

Using the PLCopen function blocks of the AsOpcUac library, it is possible to program the establishment of a connection to an OPC UA server or the reading/writing of nodes.

The AsOpcUac library can be used to exchange the process data on a B&R controller with OPC UA servers on systems of different platforms. The library's function blocks for OPC UA client functionality were developed in cooperation between OPC Foundation and PLCopen working groups. A sample project for using the function examples is installed with Automation Studio.

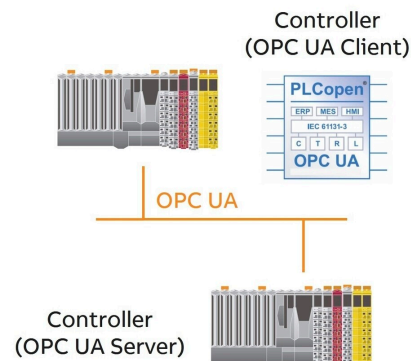


Figure 68: PLCopen client function blocks

For information about how the function blocks work and how to use them, see Automation Help:



[Programming / Libraries / Communication / AsOpcUac](#)
[Automation Software / Example programs / Communication with OPC UA client function blocks](#)

Exercise: Read/Write variables with PLCopen function blocks in the sample project

In this exercise, the functionality of libraries "AsOpcUac" and "AsOpcUas" is explained using the sample project. The client and server functionality are programmed in this project. By enabling the function blocks in the correct order, server variables can be read and written to.



In this example project, port 4841 is used for OPC UA.

- 1) Unzip file "OpcUa_Sample.zip" under path C:\BrAutomation\AS412\Samples.
- 2) Open the project in the unzipped folder and transfer it to the simulation.
- 3) Open the Watch window of the "Client01" program in the "Client" package and add all variables.
- 4) Open the Watch window of program "ServerTask" in the "Server" package and add all variables to the server variables overview.
- 5) Set variable "ExecuteConnect_0" to TRUE in order to connect to the server.
- 6) Set variable "ExecuteGetnamespaceindex_0" to TRUE to read the index of the namespace.
- 7) Set variable "ExecuteNodeGetHandle_0" to TRUE to receive the handle for the configured node.
- 8) Set variable "ExecuteRead_0" to TRUE to read server variable "VarX". This is saved in variable "VarA" on the client.
- 9) Set variable "ExecuteWrite_0" to TRUE to write client variable "VarB" to server variable "VarX".



[Programming \ Libraries \ Communication \ AsOpcUac \ Program examples](#)
[Communication \ Fieldbuses \ X20 fieldbuses / X67 system \ Bus controller \ OPC UA \ Program examples](#)

9 Methods

Methods allow a client to start a "program" on a server. The client can transfer input arguments to the server. When the server has finished processing the method, it can then transfer output arguments to the client.

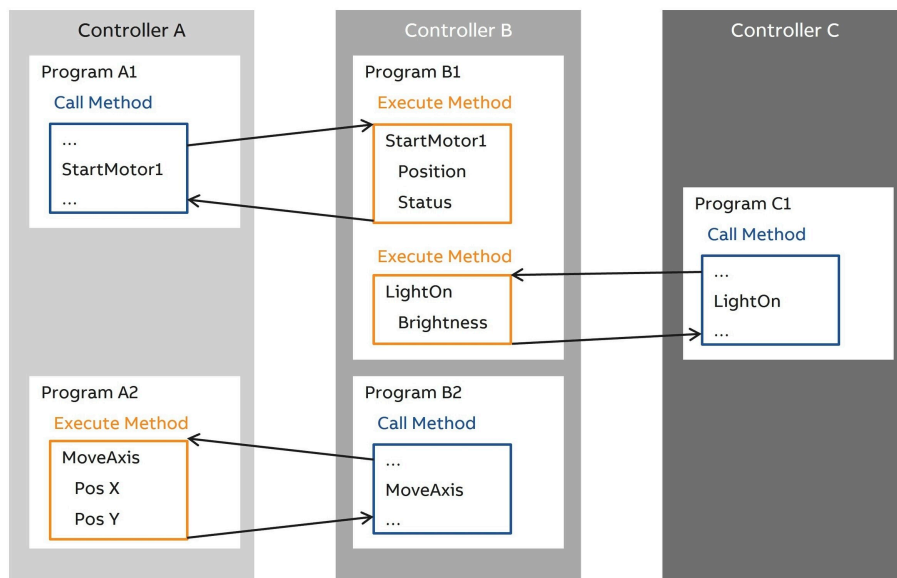


Figure 69: Example of how methods work

The advantage of OPC UA is that this handshake is already implemented. The user only has to program the method and decide when the method should be called.



[Communication \ Fieldbuses / Fieldbuses in X20 / X67 system \ Bus controller \ OPC UA \ Methods](#)
[Communication \ OPC UA \ Use cases \ Methods](#)

Exercise: Call methods on bus controller X20BC008U

Method "Reboot" is used to restart the bus controller from UaExpert.

- 1) Establish the connection to X20BC008U using UaExpert.
- 2) Navigate to Root/Objects/DeviceSet/X20BC008U/MethodSet in "Address space".
- 3) Execute the "Reboot" method via a right click on "Call".
- 4) Observe the startup of the bus controller.



All methods are executed with a right-click on "Call".

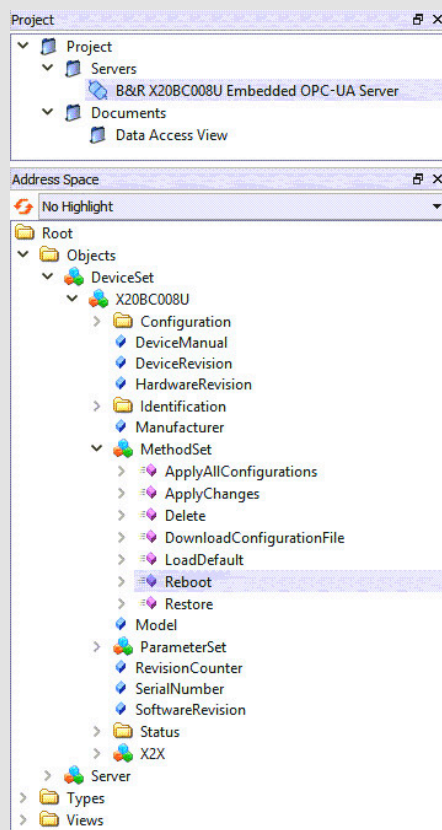


Figure 70: Method "Reboot"

Exercise: Call methods with PLCOpen function blocks in the sample project

In this exercise, a method is called that multiplies two numbers. To do this, the server must first be prepared for method calls. The prepared method can be called via program "Client03" or alternatively via UaExpert.

- 1) Open "OpcUa_Sample" and transfer it to the simulation.
- 2) Open the Watch window of program "SrvMethod" in package "Server" to add all variables.
- 3) Set variable "Step" to 1 to prepare the server for the following method calls.
- 4) Use UaExpert to establish a connection to the simulation's Automation Runtime server.
- 5) Navigate to path Root/Objects/PLC/Modules/<Default>/SrvMethod in "Address space" and call method "Multiply".
- 6) Open the Watch window of program "Client03" in package "Client" and add all variables.
- 7) Set variable "ExecuteConnect_0" to TRUE in order to connect to the server.
- 8) Set variable "ExecuteGetnamespaceindex_0" to TRUE to read the index of the namespace.
- 9) Set variable "ExecuteMethodGetHandle_0" to TRUE to receive the handle for the configured method.
- 10) Set "variable" ExecuteMethodCall_0" to TRUE to call the method.



After calling the method with UaExpert, the result is displayed if the method was processed successfully.

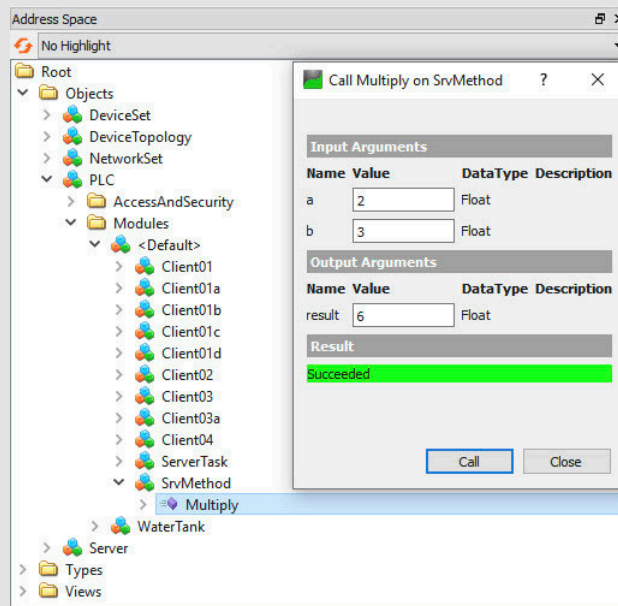


Figure 71: Result: Method "Multiply"

Exercise: Create methods

The objective of this exercise is to create and call a method that subtracts two values.

- 1) In the "Server" package, add a new "Structured Text All in One Program" with the name "MyMethod".
- 2) Copy the complete code of the "SrvMethod" task to "MyMethod".
- 3) Copy all variables of the "SrvMethod" task to "MyMethod".
- 4) Add a new OPC UA method declaration file from the Toolbox to program "MyMethod".

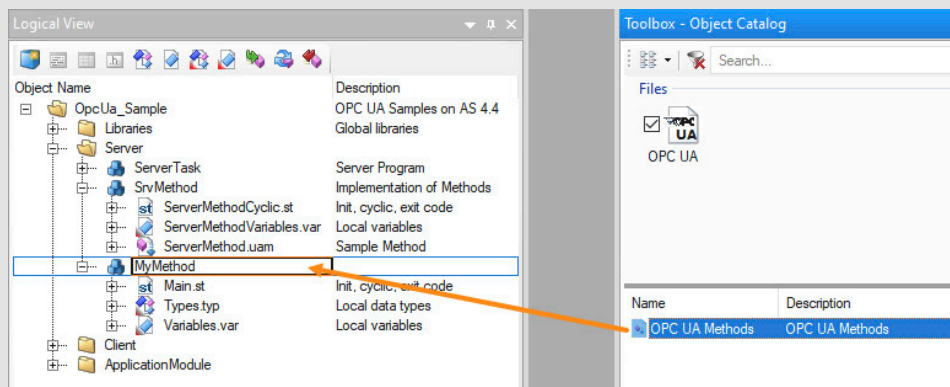


Figure 72: Adding OPC UA method declaration file

- 5) Add a new method named "Sub" to the declaration file.
- 6) Arguments "x" (direction VAR_INPUT, linked variable "VarA"), "y" (direction VAR_INPUT, linked variable "VarB") and "z" (direction VAR_OUTPUT, linked variable "VarResult") are created in this method.

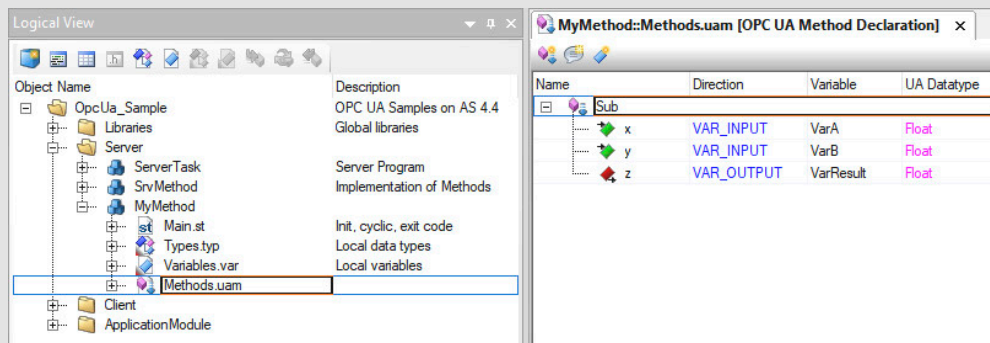


Figure 73: OPC UA method declaration file: Variables

- 7) Enable method "Sub" in the OPC UA default View in task "MyMethod" and delete example "WaterTank".
- 8) Change the "MethodName" in the program code of "MyMethod" from "Multiply" to "Sub" at steps 1 (prepare UaSrv_MethodOperate) and 4 (start operate).
- 9) In step 3, change the calculation to "VarResult:= VarA - VarB" execute method code).
- 10) Transfer the project.
- 11) Set variable "Step" to 1 to prepare the server for the following method calls.
- 12) Call methods with UaExpert.

10 Appendix

10.1 OPC UA FX package

 Enabling or disabling the OPC UA FX Technology Package as well as changing the version of the Technology Package may cause the controller to restart during project installation.

 [Communication \ OPC UA FX \ General \ Installing the Technology Package](#)

Optional exercise: Install OPC UA FX package

To be able to use the OPC UA FX Technology Package, it must be installed in Automation Studio via the Upgrades dialog box.

- 1) Under Extras \ Upgrades, download and install the "OpcUaFx" Technology Package.
- 2) Select the OPC UA FX Technology Package under Project \ Change runtime versions.

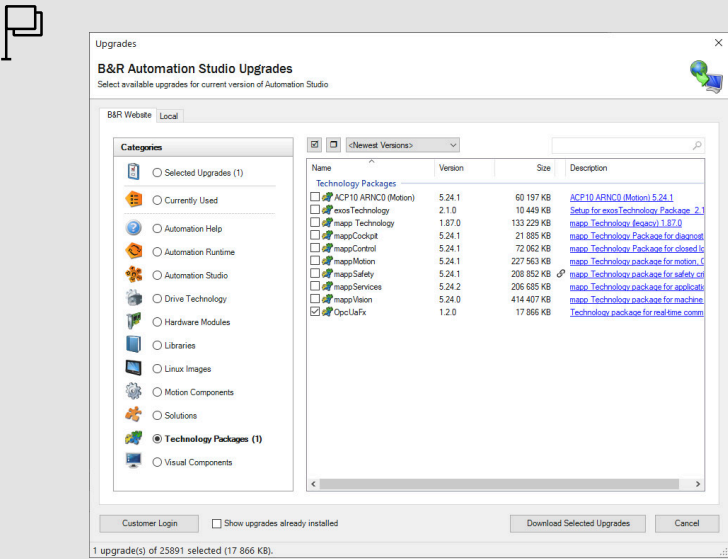


Figure 74: Extras\Upgrades

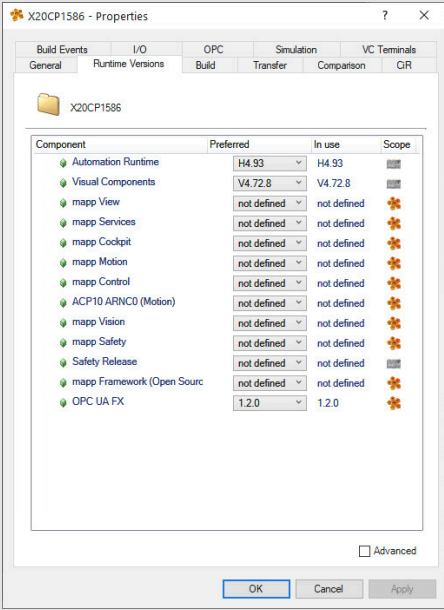


Figure 75: Project \ Change Runtime versions

10.2 Firmware update X20BC008U



If an error occur during the firmware download, we recommend using a different browser.



[Hardware \ X20 System \ X20 module \ Bus controller \ X20BC008U \ Firmware update](#)

Optional exercise: Perform a firmware update for X20BC008U

The following steps are required to update the firmware of the bus controller:

- 1) Download the firmware from the B&R website (Search term: "Firmware X20BC008U").
- 2) Call up the IP address or the host name of the bus controller in the browser (e.g. <http://192.168.1.2> or <http://bropc2>).
- 3) Select the downloaded firmware file under "Advanced/Firmware download".
- 4) Select "Start download".
- 5) When the download is complete, restart the bus controller via "Restart bus controller".



After the downloaded firmware has been selected and command "Start download" has been executed, the firmware is updated. After download, the bus controller must be restarted.

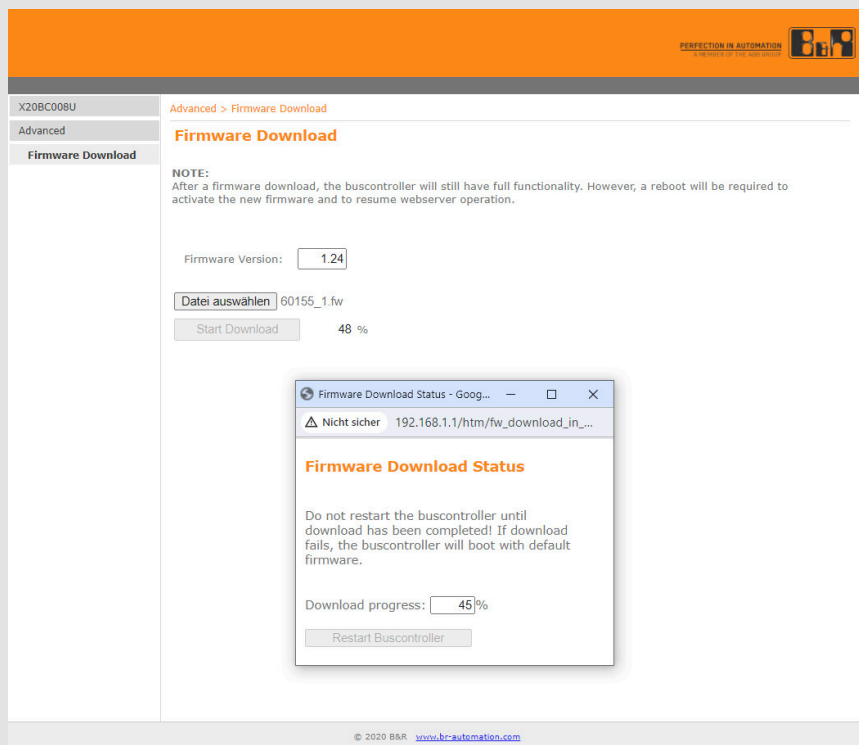


Figure 76: Firmware download

10.3 Optional exercises

Optional exercise: Publisher / Subscriber continuation

The objective of the exercise is to make a variable available to the subscriber. The program from a previous exercise is used for this ("Exercise: Create a program and initialize variables" on page 26). The variable "Distance" is transferred.



Figure 77: Example of a possible topology



In order to communicate with another network station via ARsim, the IP address of the ETH adapter of the PC running ARsim must be entered in the ARsim Ethernet configuration, not 127.0.0.1.



[Communication \ OPC UA FX \ PubSub \ Diagnostics \ FAQ](#)

- 1) In the Configuration View under Connectivity \ OpcUaFx, add the PublishSubscribe file (UaPubSub) via the Toolbox.
- 2) In the PublishSubscribe file under WriterGroups, set the publishing interval to 100 ms.
- 3) In the PublishSubscribe file under WriterGroups \ PublishedDataSet, select the variable "Distance" in the source variable field.
- 4) Transfer the project.
- 5) Check the Logger; an entry such as "OPC UA PubSub was started successfully" should be present.
- 6) In the PublishSubscribe file, right-click on PubsubConnection and select the option "Export all published DataSets for all connections to *.uabinary".
- 7) Save the .uabinary file.
- 8) Add a new configuration in the Configuration View and name it "Subscriber" (hardware for example PC = simulation).
- 9) Enable the OPC UA system in the CPU configuration.
- 10) Add a new ST program with the name "Subscriber" in the Logical View.

```
VAR
    DistanceSubscriber : USINT;
END_VAR

PROGRAM _INIT
    DistanceSubscriber := DistanceSubscriber;
END_PROGRAM
```

- 11) Add the OPC UA default view in the Configuration View under Connectivity\OpcUA using the Toolbox.
- 12) Enable the variable "DistanceSubscriber".

- 13) In the Configuration View under Connectivity \ OpcUaFx, add the PublishSubscribe file (UaPubSub) via the Toolbox.
- 14) In the PublishSubscribe file, right-click on ReaderGroups and click on the option "Import DataSets from *.uabinary".
- 15) Select the previously saved .uabinary file.
- 16) In the PublishSubscribe file under ReaderGroups \ SubscribedDataSet, select variable "DistanceSubscriber" in the target variable field.
- 17) Delete WriterGroup1 in the PublishSubscribe file.
- 18) In the Logical View, adjust the Ethernet settings so the local laptop IP address is used for the simulation, for example.
- 19) In the software configuration, assign the "Subscriber" task to task class #1 (100 ms).
- 20) Transfer the project, e.g. offline installation to the simulation.
- 21) In UaExpert, set the "Switch" variable to TRUE and check whether the "Distance" variable changes its value cyclically.
- 22) Use Monitor mode in Automation Studio to check the "DistanceSubscriber" variable to see whether it changes its value cyclically.

11 Summary

OPC Unified Architecture (OPC UA) is a manufacturer-independent communication protocol for automation applications in industry. It enables end-to-end communication from individual sensors and actuators to the ERP system or the cloud.

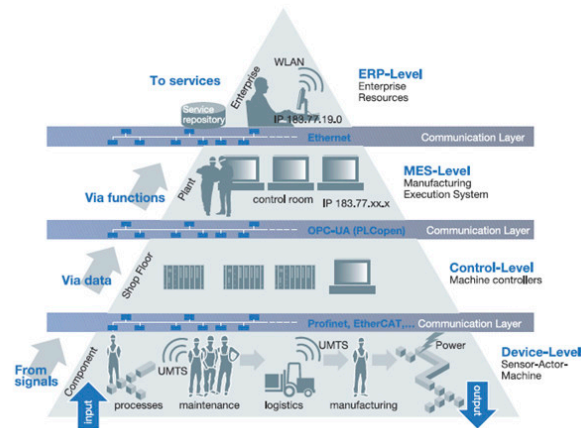


Figure 78: OPC UA pyramid

Configuring OPC UA devices and assigning access rights in B&R's Automation Studio engineering environment is completed with just a few clicks. PLCopen function blocks are used to implement OPC UA functions in an automation project.

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B&R
Industrial Automation GmbH
A member of the ABB Group
B&R Straße 1
5142 Eggelsberg, Austria
office@br-automation.com

t +43 7748 6586-0
f +43 7748 6586-26

br-automation.com

